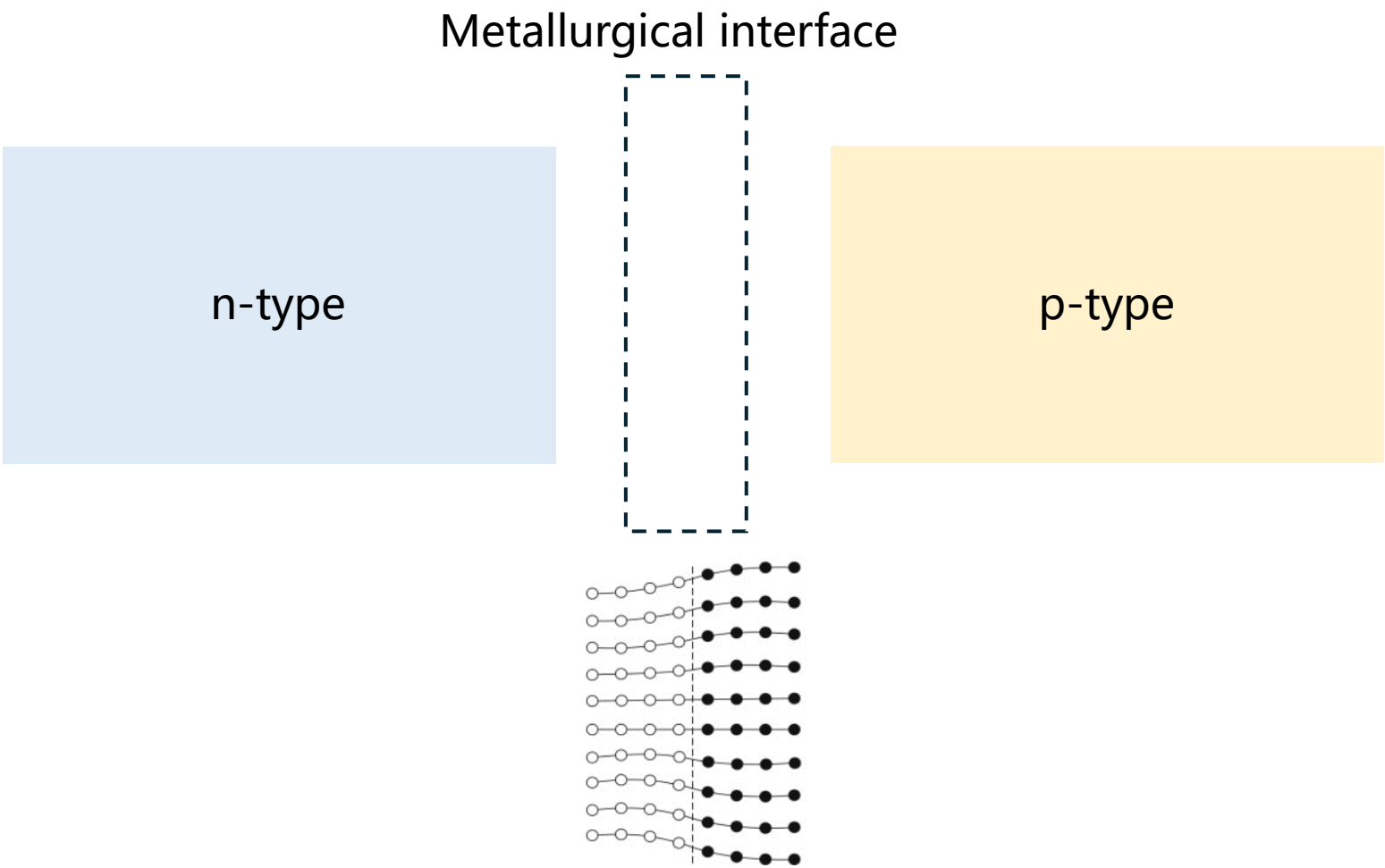
Chapter 7. pn junction

Previously on what you have learned in semiconductor physics:

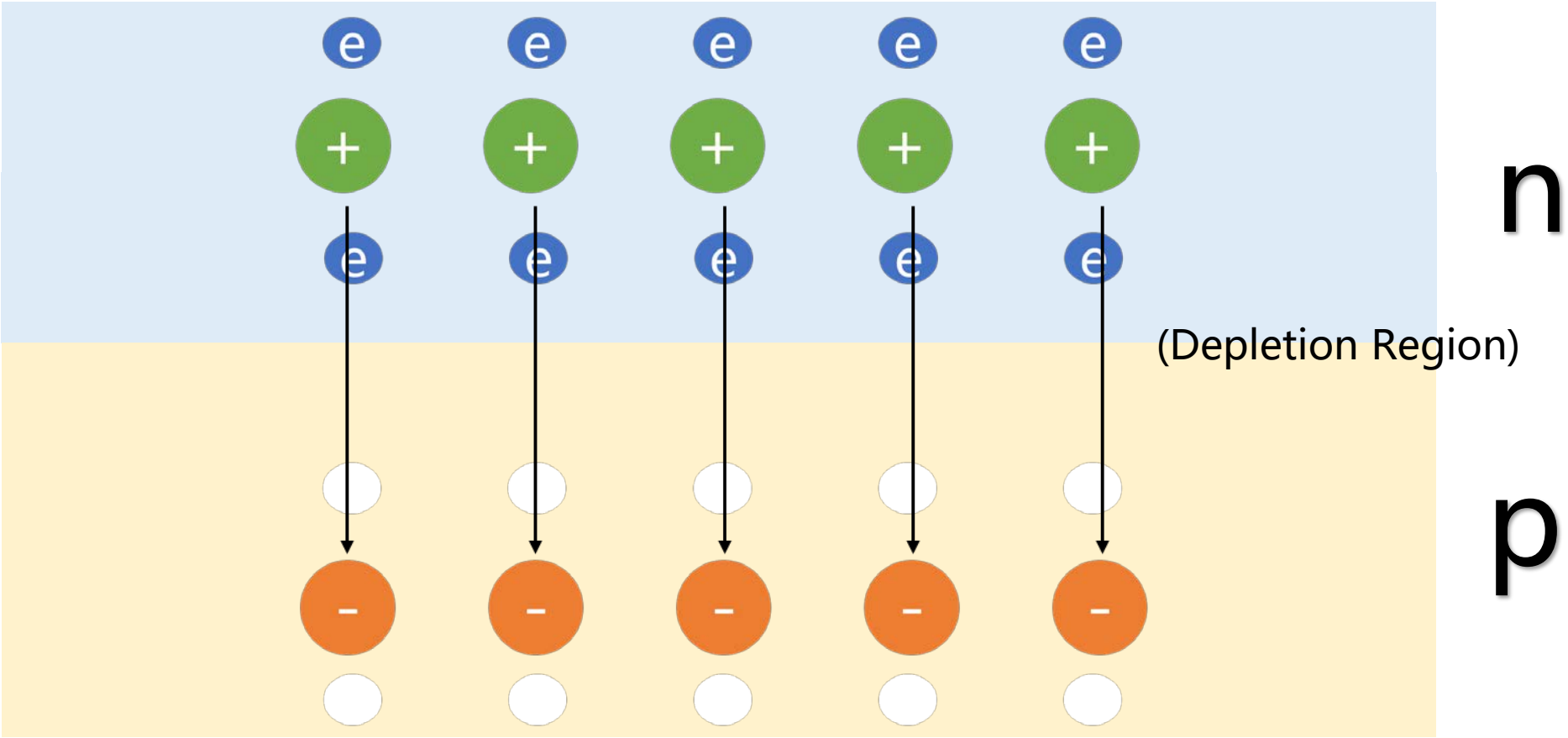
1. Knowledge of electron orbitals and energy levels.
2. Valence band, conduction band, and forbidden energy gap.
3. Chemical potential (Fermi level) and its significance.
4. Introduction of impurities into the semiconductor material.
5. How n-type and p-type regions are created.

pn junction: concept

If n- and p-type uniformly doped semiconductors are tightly bonded together：

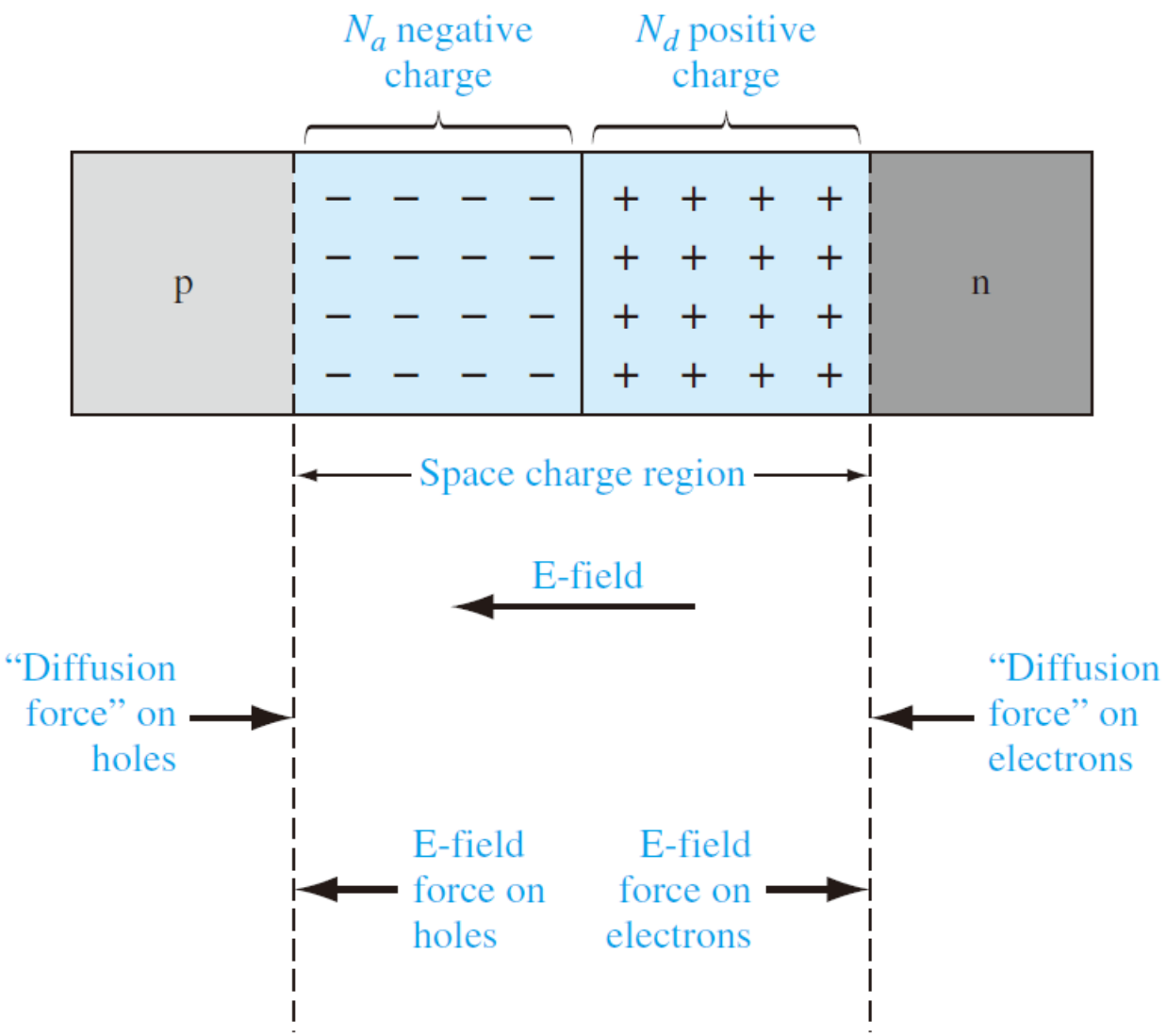


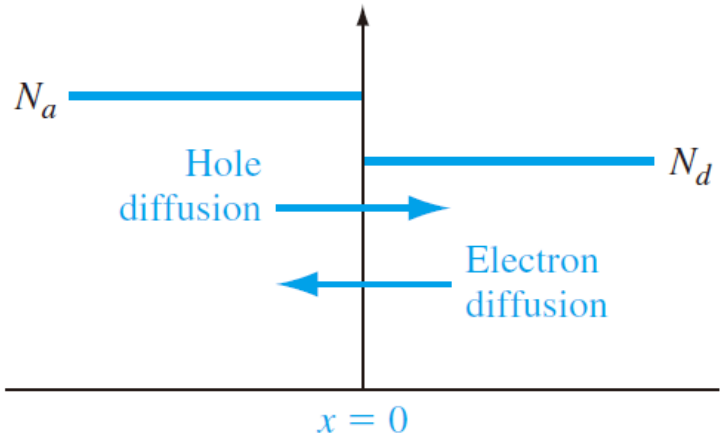
pn junction: concept

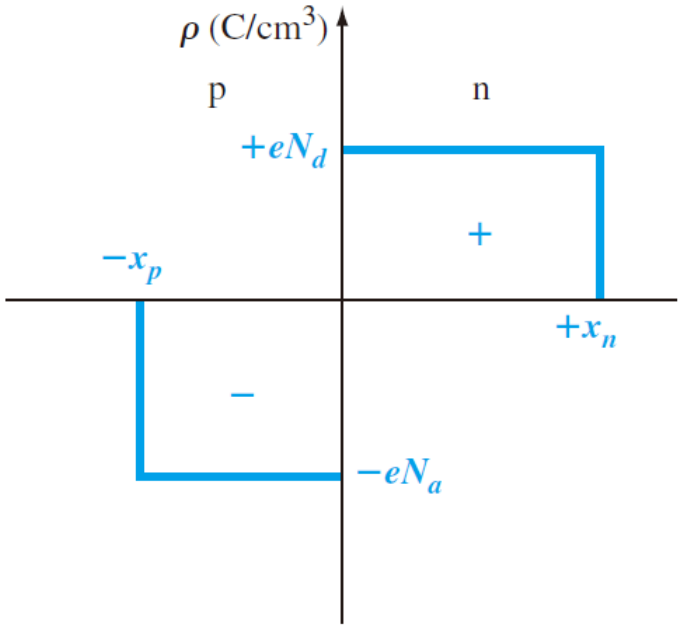


Depletion refers to that the free carriers are depleted, this region is also called space charge region

pn junction: charge density



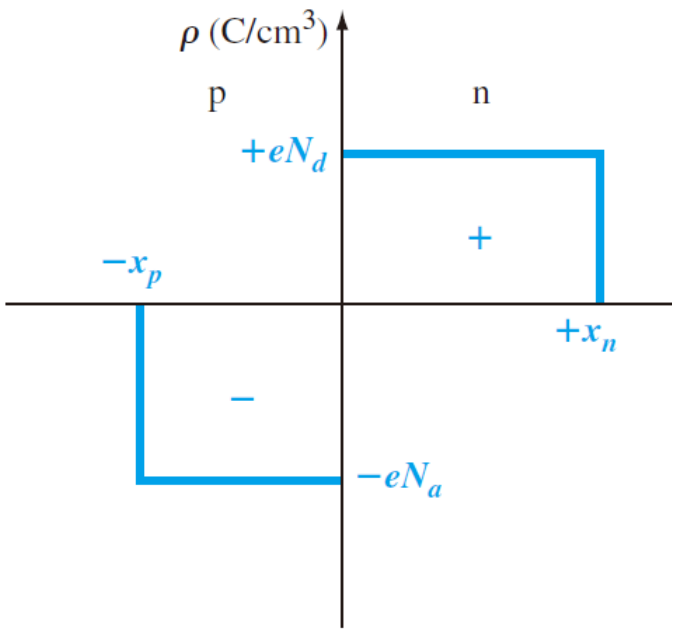




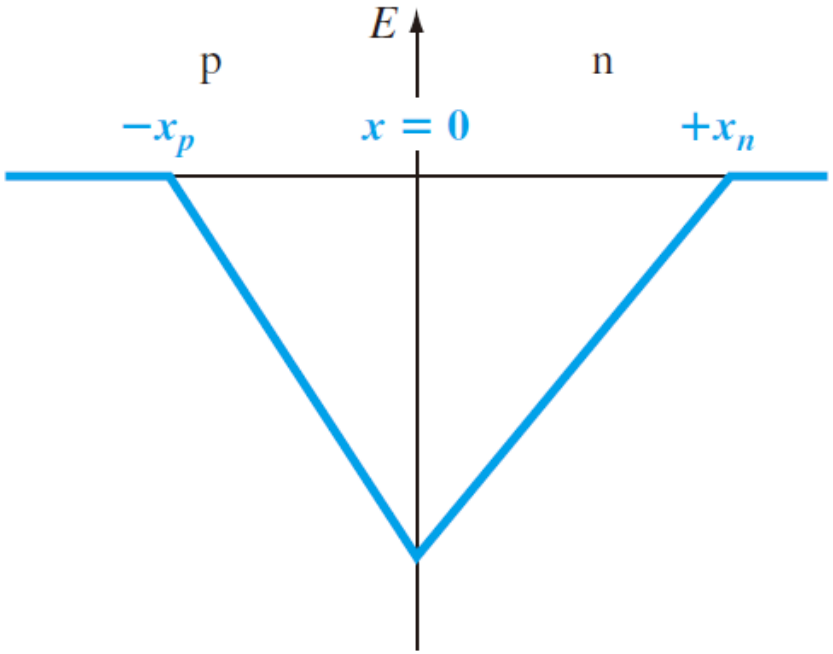
Space charge density

pn junction: field and potential

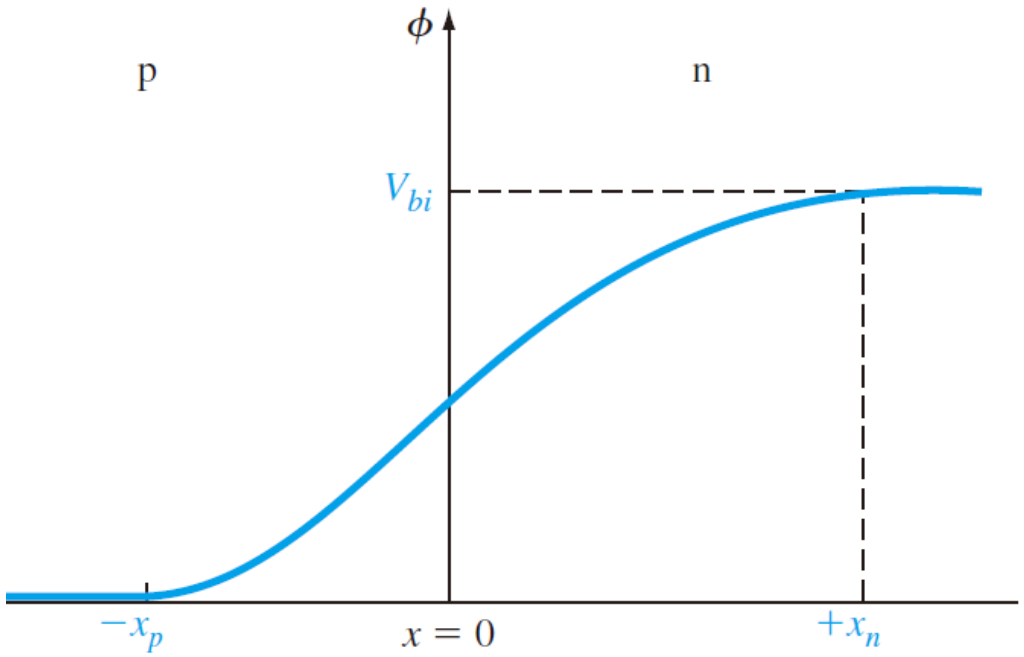
Poisson' s equation: $$\frac{d^2\phi(x)}{dx^2} = \frac{-\rho(x)}{\epsilon_s} = -\frac{d\mathrm{E}(x)}{dx}$$



Space charge density



Electric field

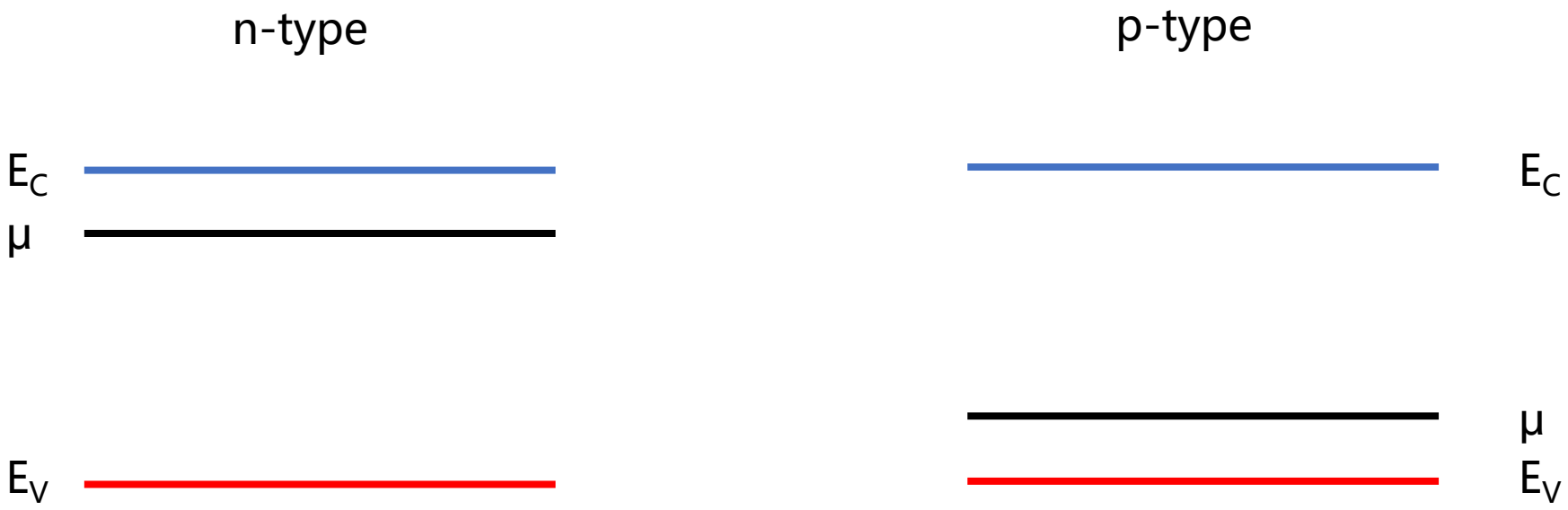


Potential

Discussion: What are the boundary conditions of the corresponding equations?

pn junction on the energy perspective

In energy band theory, the energy refers to electron potential $e\phi(x)$



pn junction on the energy perspective

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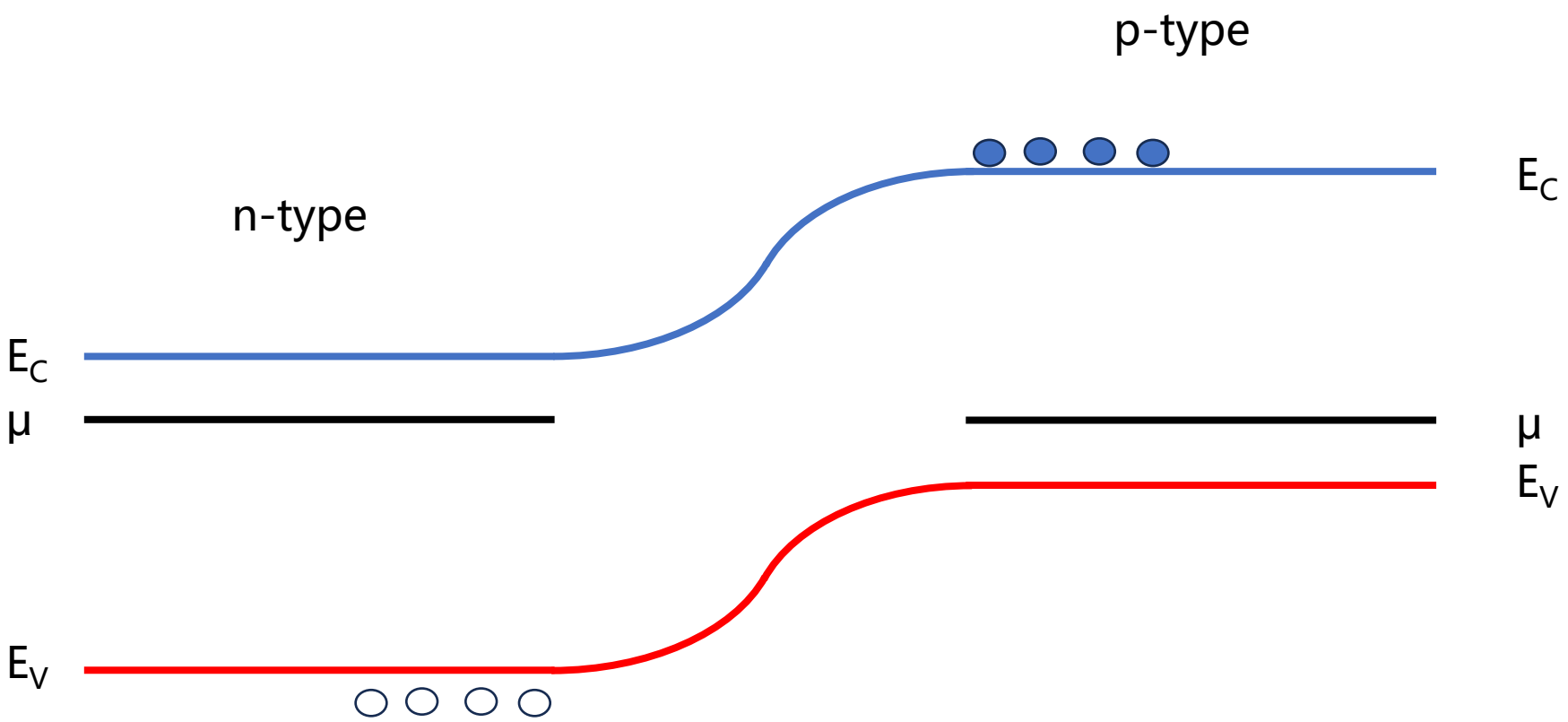
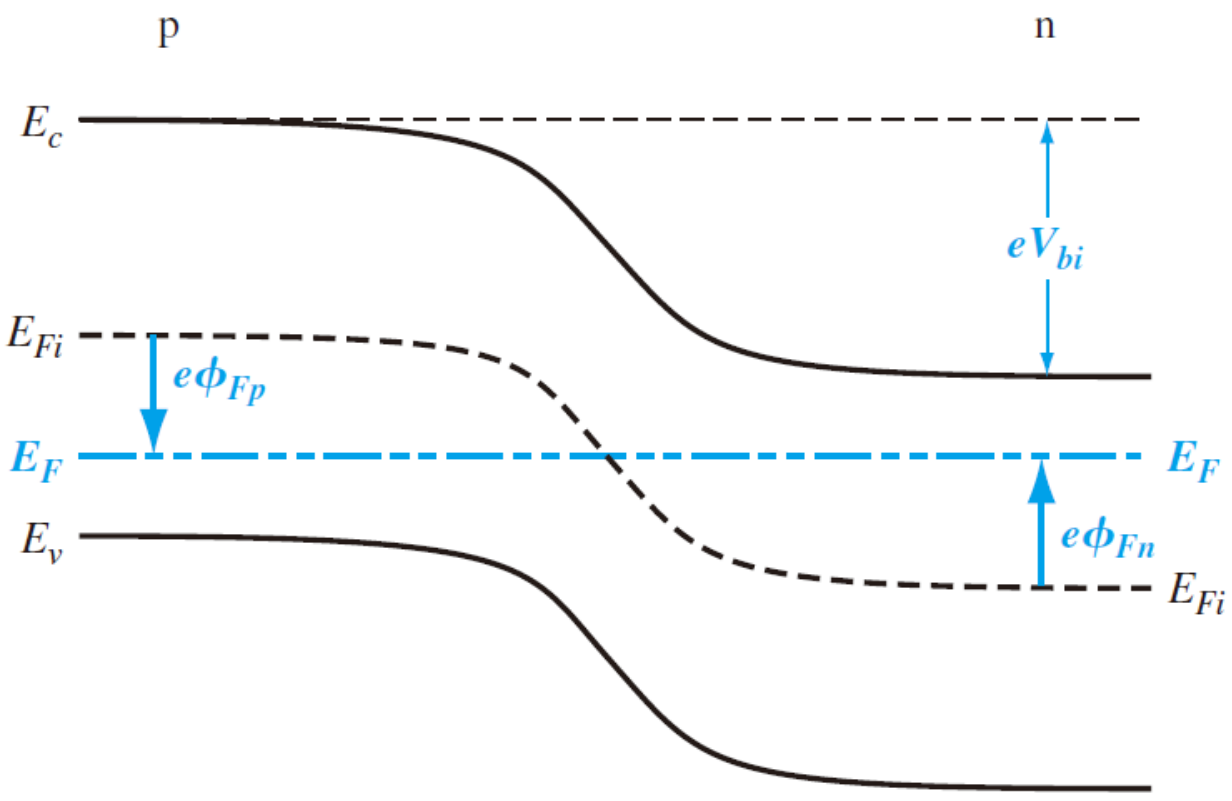


The physics of pn junction: build-in potential V_{bi}



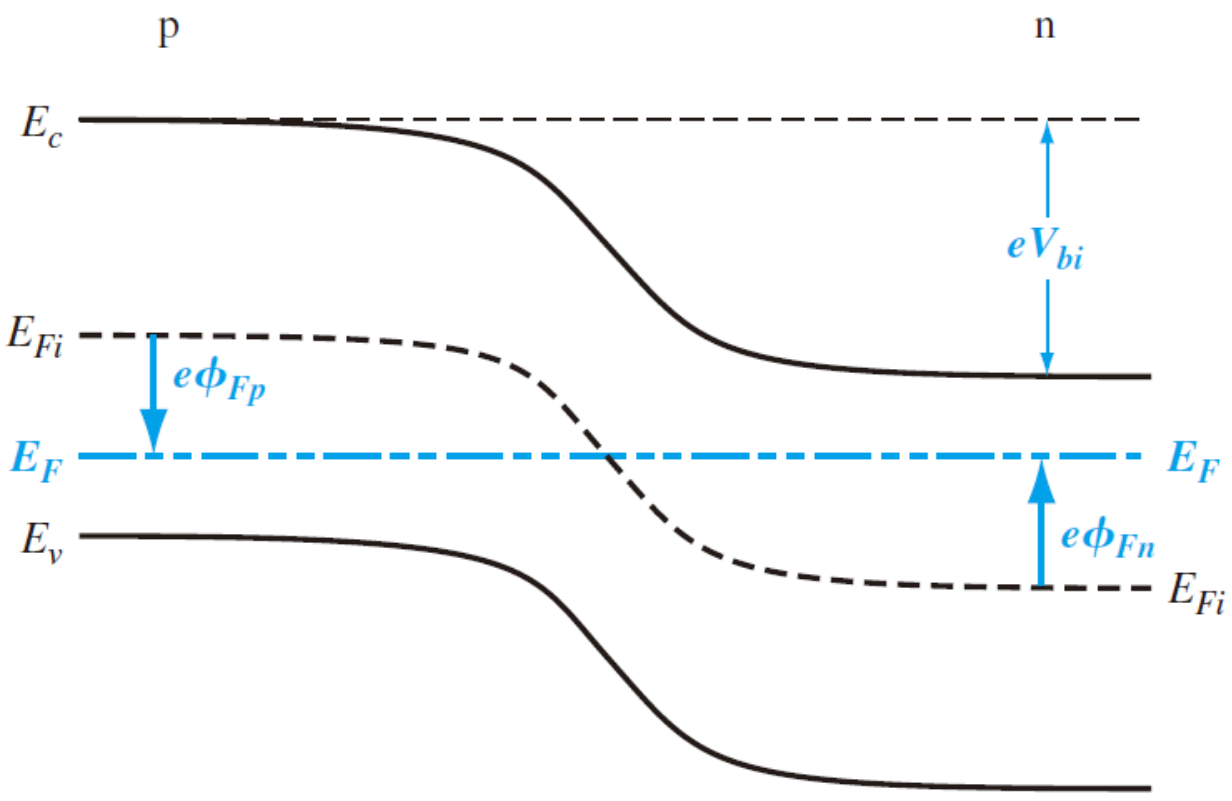
Two assumptions:

1. Carriers follow Boltzmann approximation, which means E_F is far enough from E_c or E_v.
2. The doped impurities are completely ionized, which means the temperature is sufficiently high.

Discussion:

What will happen if Assumption 1 or Assumption 2 is not true, respectively?

The physics of pn junction: build-in potential V_{bi}



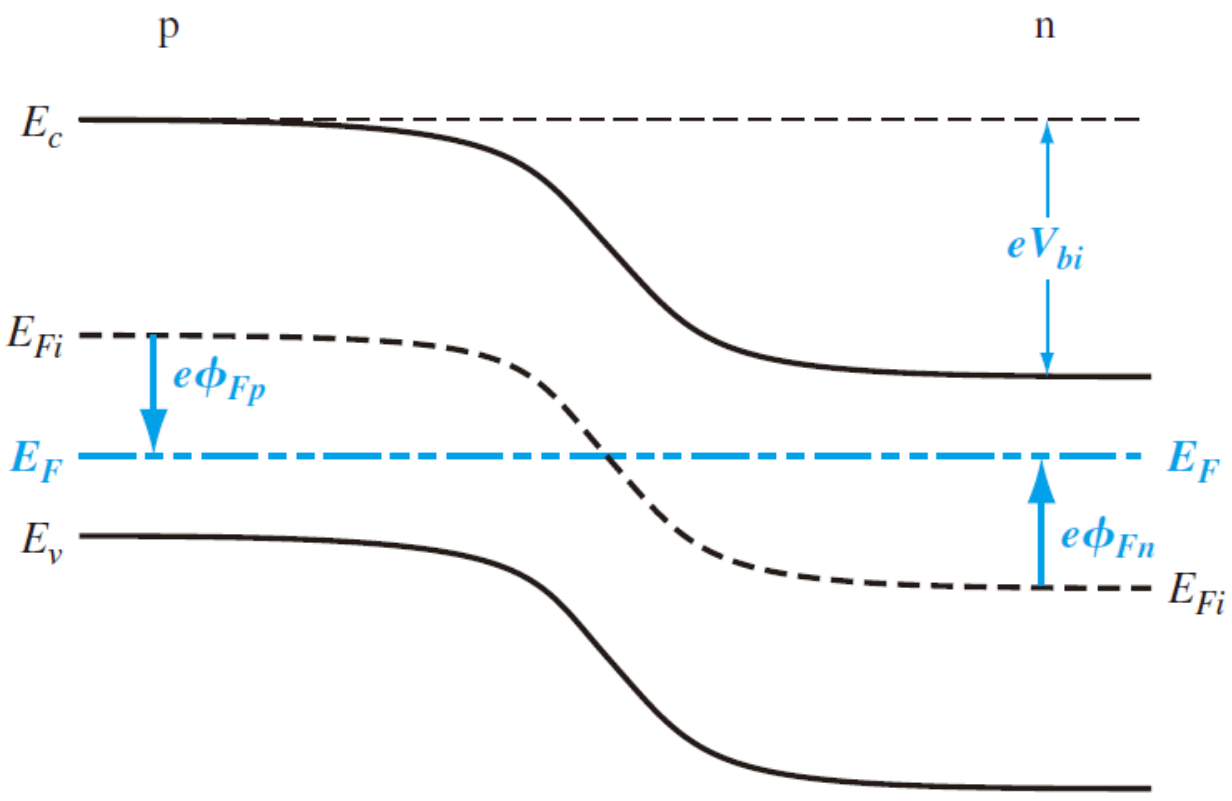
V_{bi} equals to the sum of the relative shifts of the energy bands of the materials on both sides of the pn junction:

$$V_{bi} = |\phi_{Fn}| + |\phi_{Fp}|$$

$$e\phi_{Fn} = E_{Fi} - E_F \qquad e\phi_{Fp} = E_{Fi} - E_F$$

But we don’ t know E_{Fi} or E_F, what we can measure is the carrier concentration n_0 of n-type and p_0 of p-type semiconductors, respectively.

The physics of pn junction: build-in potential V_{bi}



Consider in the n- region, based on Assumption 1:

$$n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right) \quad p_0 = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

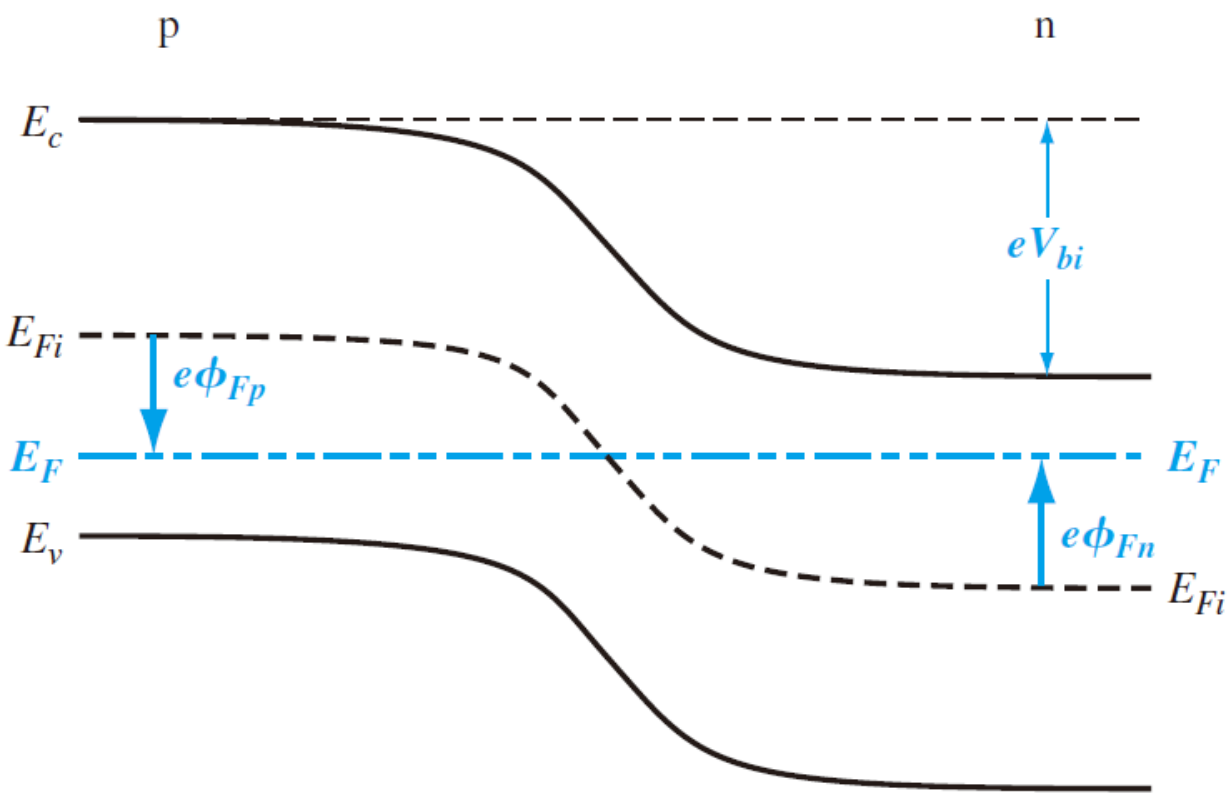
Also ,assume the net doping concentration is N_d and N_a for n- and p-type, respectively. Based on Assumption 2, we have:

$$n_0 = N_d \qquad p_0 = N_a$$

$$\phi_{Fn} = \frac{-kT}{e} \ln\left(\frac{N_d}{n_i}\right)$$

$$\phi_{Fp} = +\frac{kT}{e} \ln\left(\frac{N_a}{n_i}\right)$$

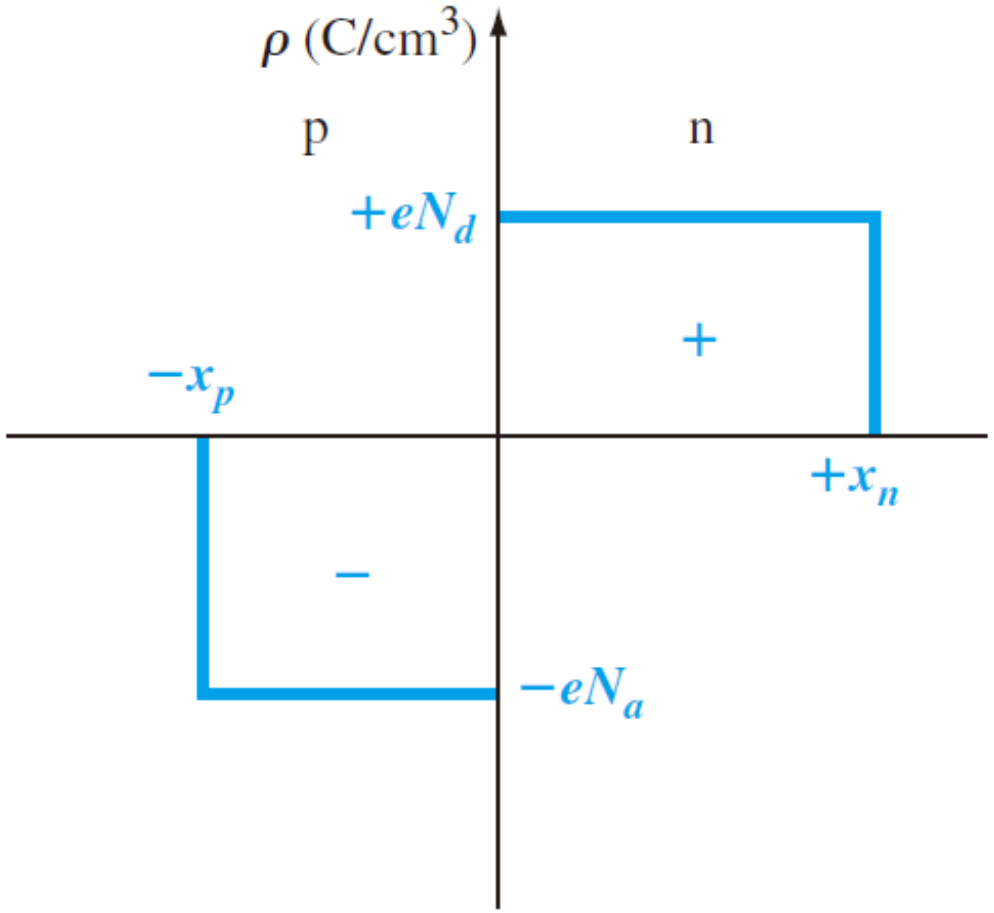
The physics of pn junction: build-in potential V_{bi}



$$V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right) = V_t \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

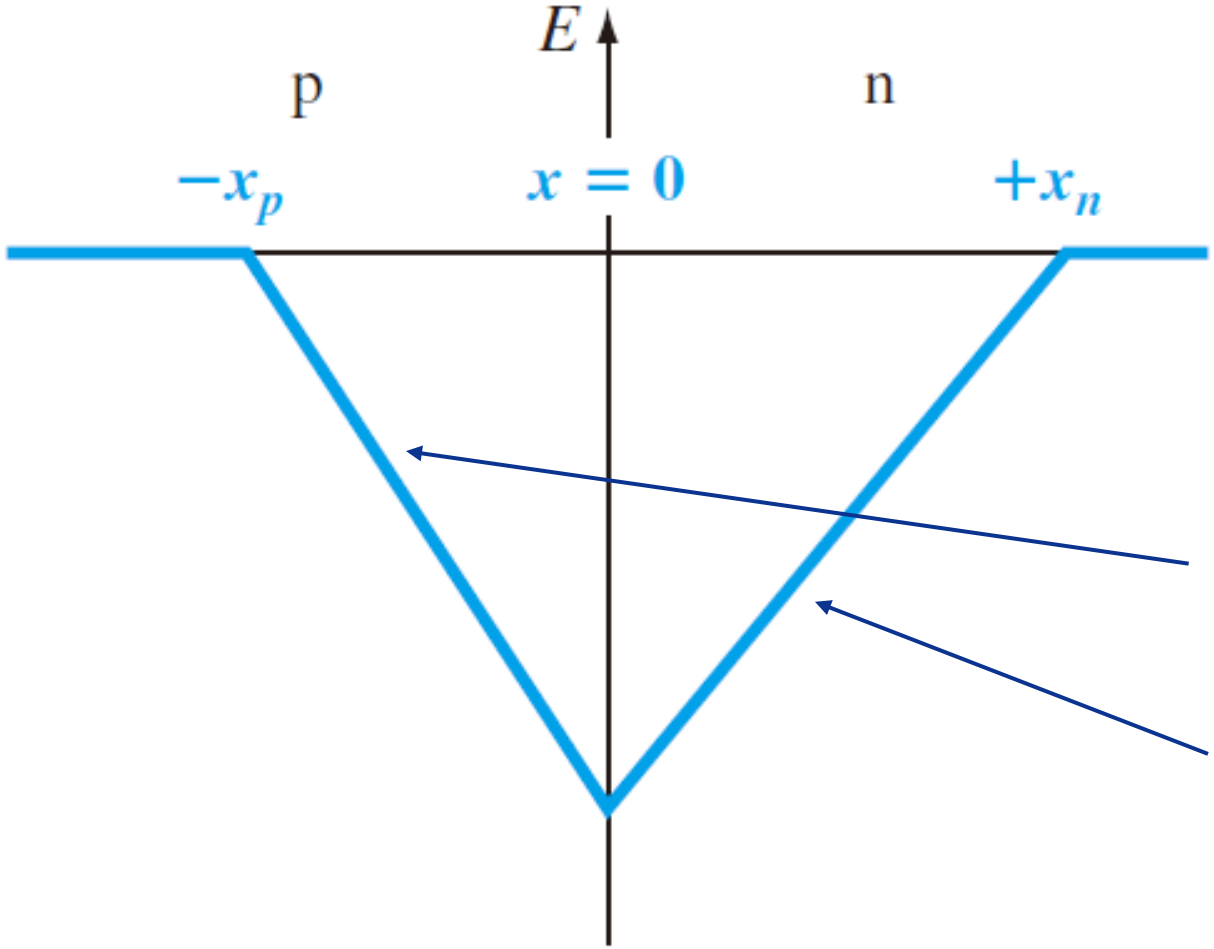
For room temperature $T = 300\ K$, $\frac{kT}{e} = 0.0259\ V$

The physics of pn junction: space charge distribution



For uniformly weak doped junction, based on assumption 2, the space charge density equals to the net doping concentration of n- and p-type semiconductors times electron charge, respectively.

The physics of pn junction: electric field



In 1-D, we have:

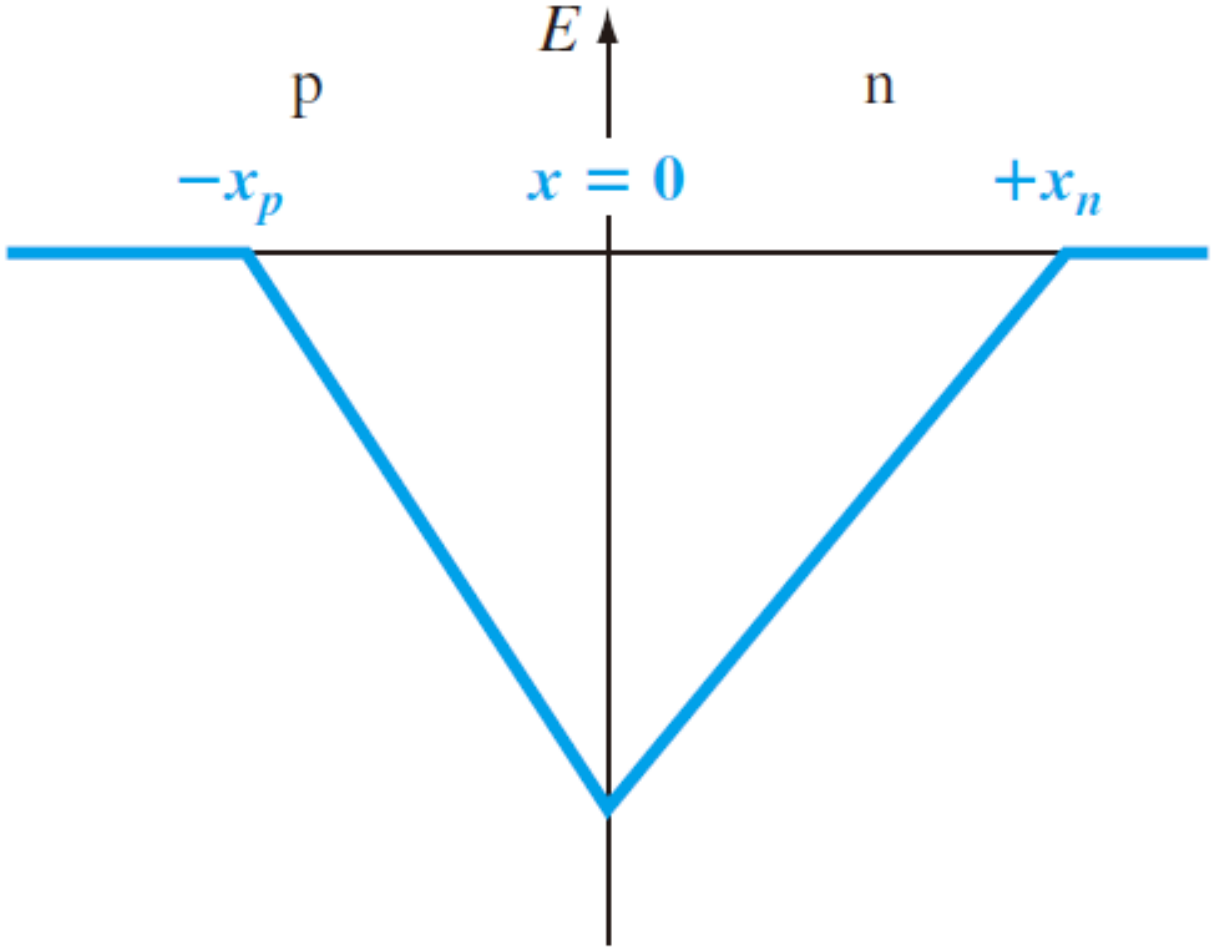
$$\frac{d^2\phi(x)}{dx^2} = \frac{-\rho(x)}{\epsilon_s} = -\frac{d\mathrm{E}(x)}{dx}$$

$$\rho(x) = \begin{cases} -eN_a & -x_p < x < 0 \\ eN_d & 0 < x < x_n \end{cases}$$

$$\mathrm{E} = \int \frac{\rho(x)}{\epsilon_s}\,dx = -\int \frac{eN_a}{\epsilon_s}\,dx = \frac{-eN_a}{\epsilon_s}x + C_1$$

$$\mathrm{E} = \int \frac{(eN_d)}{\epsilon_s}\,dx = \frac{eN_d}{\epsilon_s}x + C_2$$

The physics of pn junction: electric field



The boundary conditions are:

$$E(-x_p) = E(x_n) = 0$$

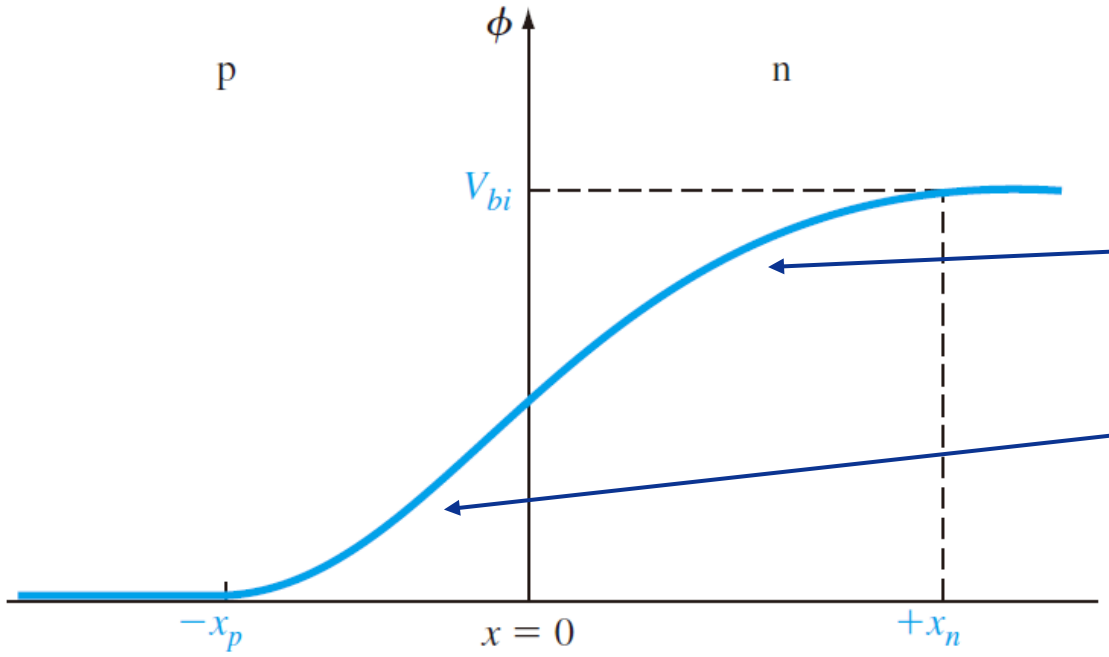
$$\mathrm{E} = \frac{-eN_a}{\epsilon_s}(x + x_p) \qquad -x_p \le x \le 0$$

$$\mathrm{E} = \frac{-eN_d}{\epsilon_s}(x_n - x) \qquad 0 \le x \le x_n$$

This tells us that the depletion region of the pn junction is a charge lever, and the width of the junction region on one side of the junction is inversely proportional to the charge density.

$$N_a x_p = N_d x_n$$

The physics of pn junction: potential



$$\frac{d^2\phi(x)}{dx^2} = \frac{-\rho(x)}{\epsilon_s} = -\frac{d\mathrm{E}(x)}{dx}$$

$$\phi(x) = \frac{eN_d}{\epsilon_s}\left(x_n \cdot x - \frac{x^2}{2}\right) + C_2'$$

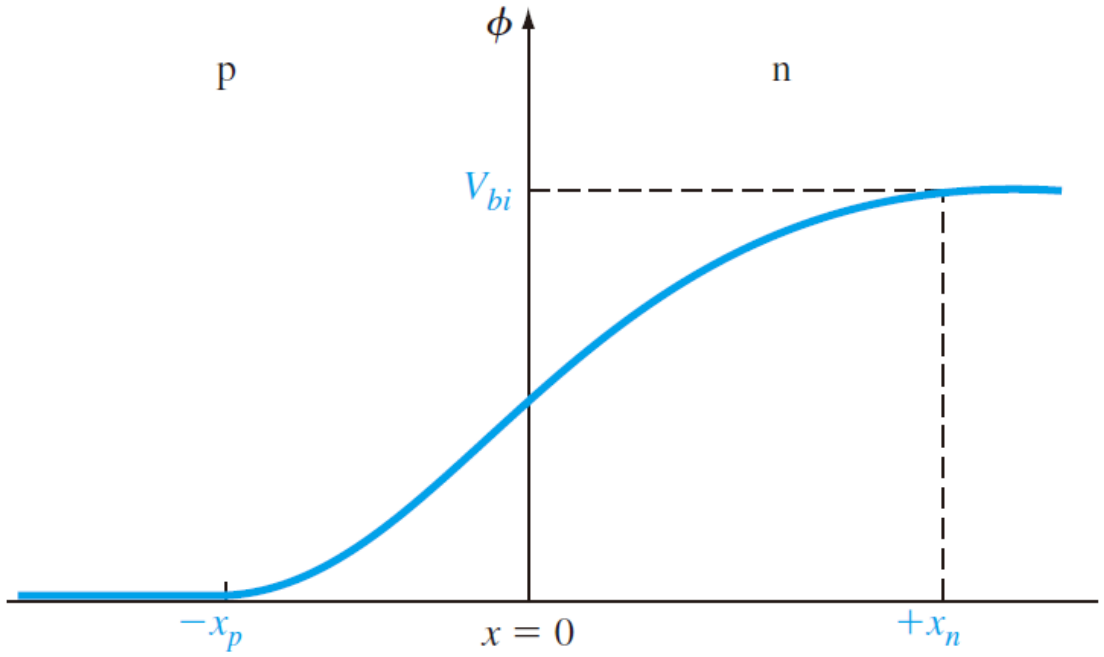
$$\phi(x) = \frac{eN_a}{\epsilon_s}\left(\frac{x^2}{2} + x_p \cdot x\right) + C_1'$$

The boundary conditions are:

$$\phi(-x_p) = 0 \qquad \phi(x \longrightarrow 0_-) = \phi(x \longrightarrow 0_+)$$

$$C_1' = \frac{eN_a}{2\epsilon_s} x_p^2 \qquad C_2' = \frac{eN_a}{2\epsilon_s} x_p^2$$

The physics of pn junction: potential



$$\phi(x) = \frac{eN_a}{2\epsilon_s}(x + x_p)^2 \qquad (-x_p \le x \le 0)$$

$$\phi(x) = \frac{eN_d}{\epsilon_s}\left(x_n \cdot x - \frac{x^2}{2}\right) + \frac{eN_a}{2\epsilon_s}x_p^2 \qquad (0 \le x \le x_n)$$

$$V_{bi} = |\phi(x = x_n)| = \frac{e}{2\epsilon_s}\left(N_d x_n^2 + N_a x_p^2\right)$$

The physics of pn junction: space charge width

Now we know how V_{bi} is determined by N_d and N_a. The next thing we' ll discuss is the width of space charge (depletion) region.

$$\begin{cases} N_a x_p = N_d x_n \\ V_{bi} = |\phi(x = x_n)| = \dfrac{e}{2\epsilon_s}\left(N_d x_n^2 + N_a x_p^2\right) \end{cases}$$

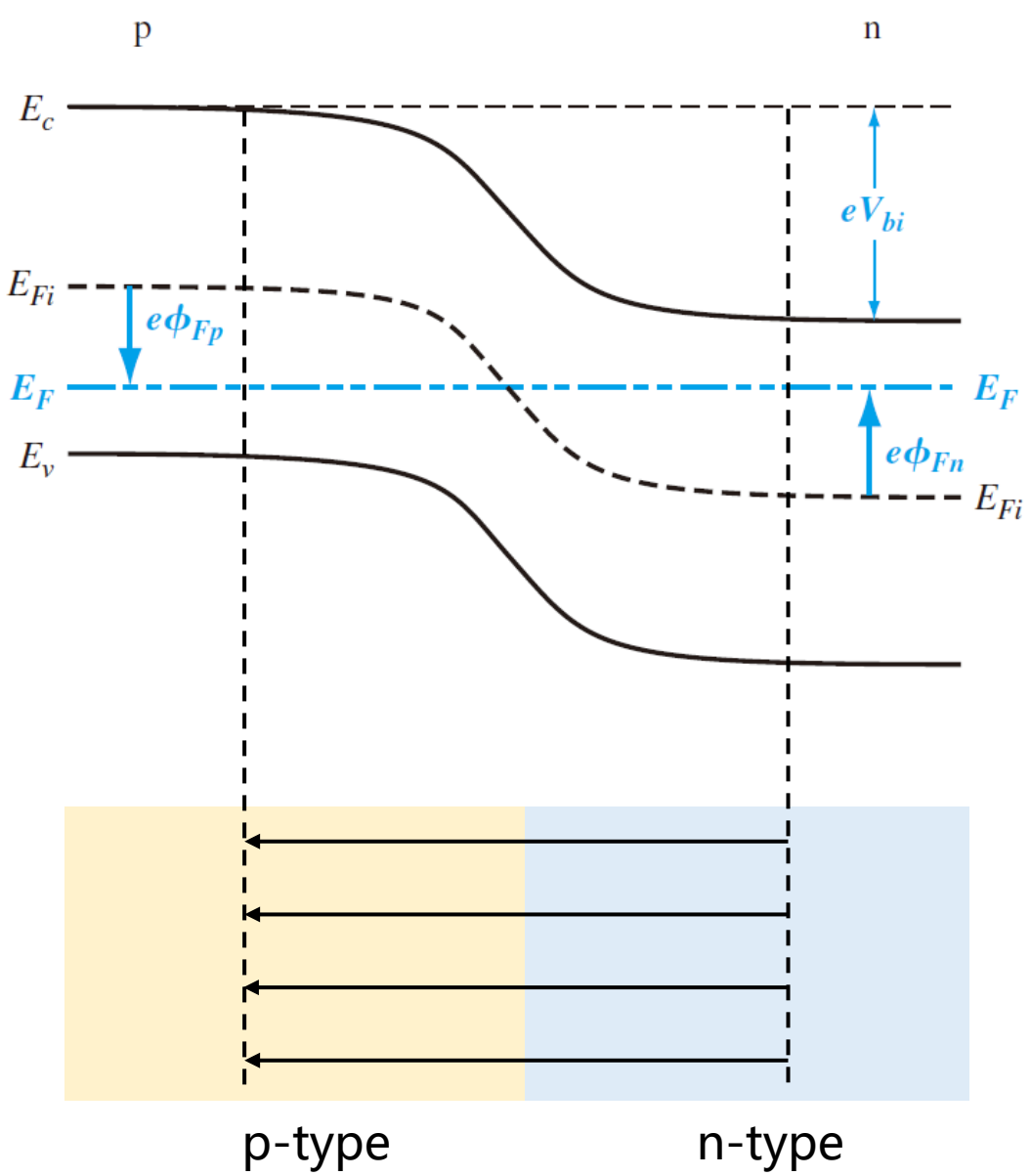
$$\Rightarrow$$

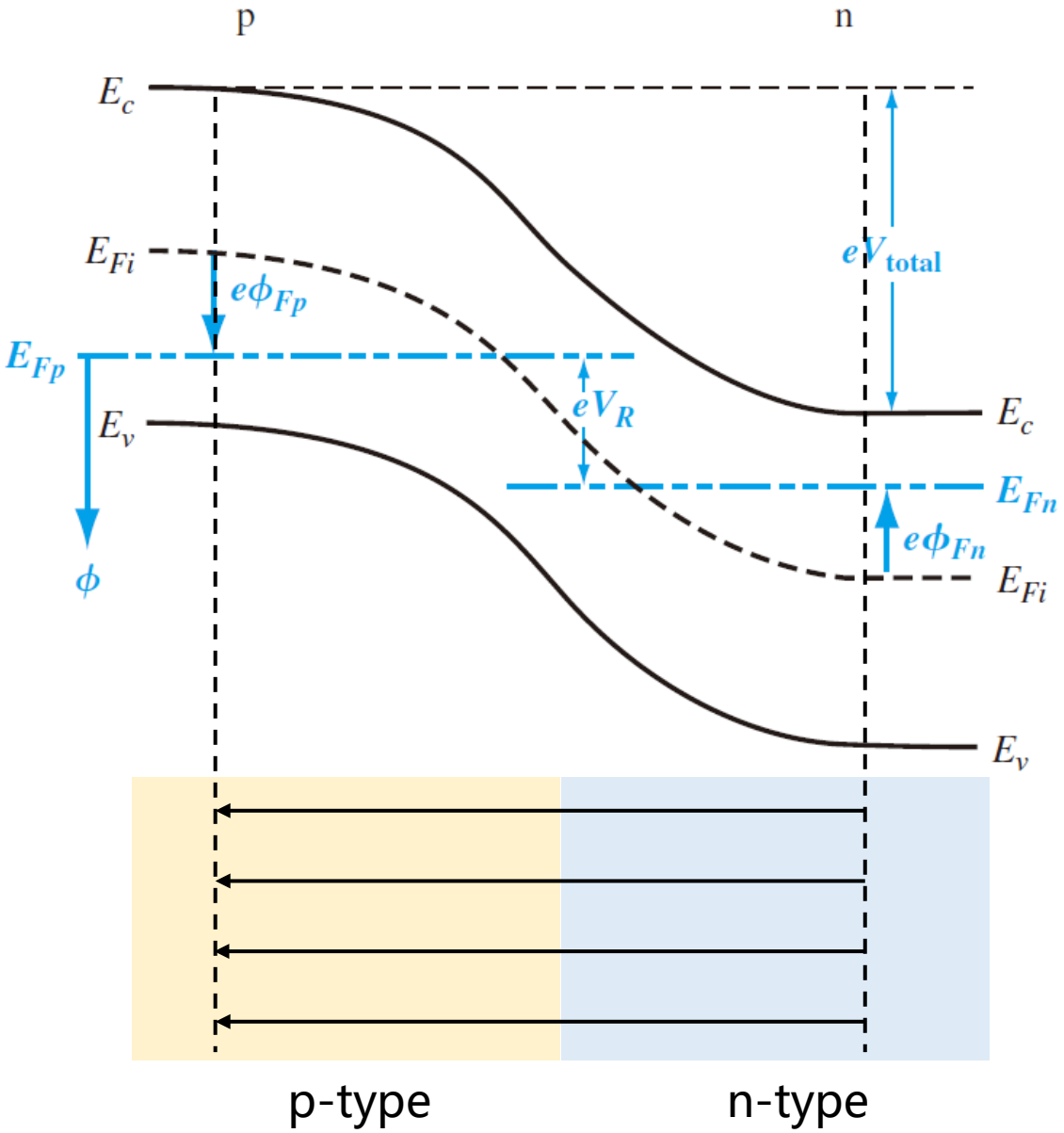
$$\boxed{x_n = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a}{N_d}\right]\left[\frac{1}{N_a + N_d}\right]\right\}^{1/2}}$$

$$\boxed{x_p = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_d}{N_a}\right]\left[\frac{1}{N_a + N_d}\right]\right\}^{1/2}}$$

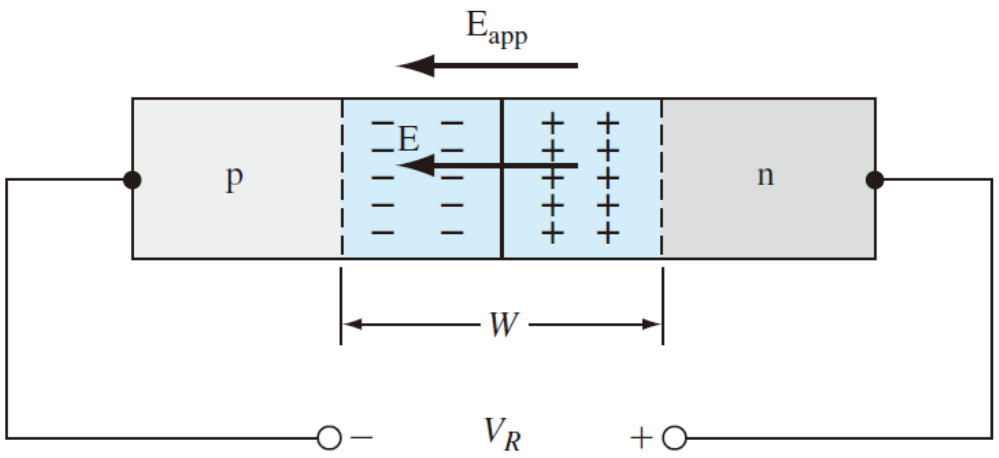
$$W = x_n + x_p = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a + N_d}{N_a N_d}\right]\right\}^{1/2}$$

The physics of pn junction: with reverse bias V_R





The physics of pn junction: with reverse bias V_R



Reverse bias extends the length of the space charge region:

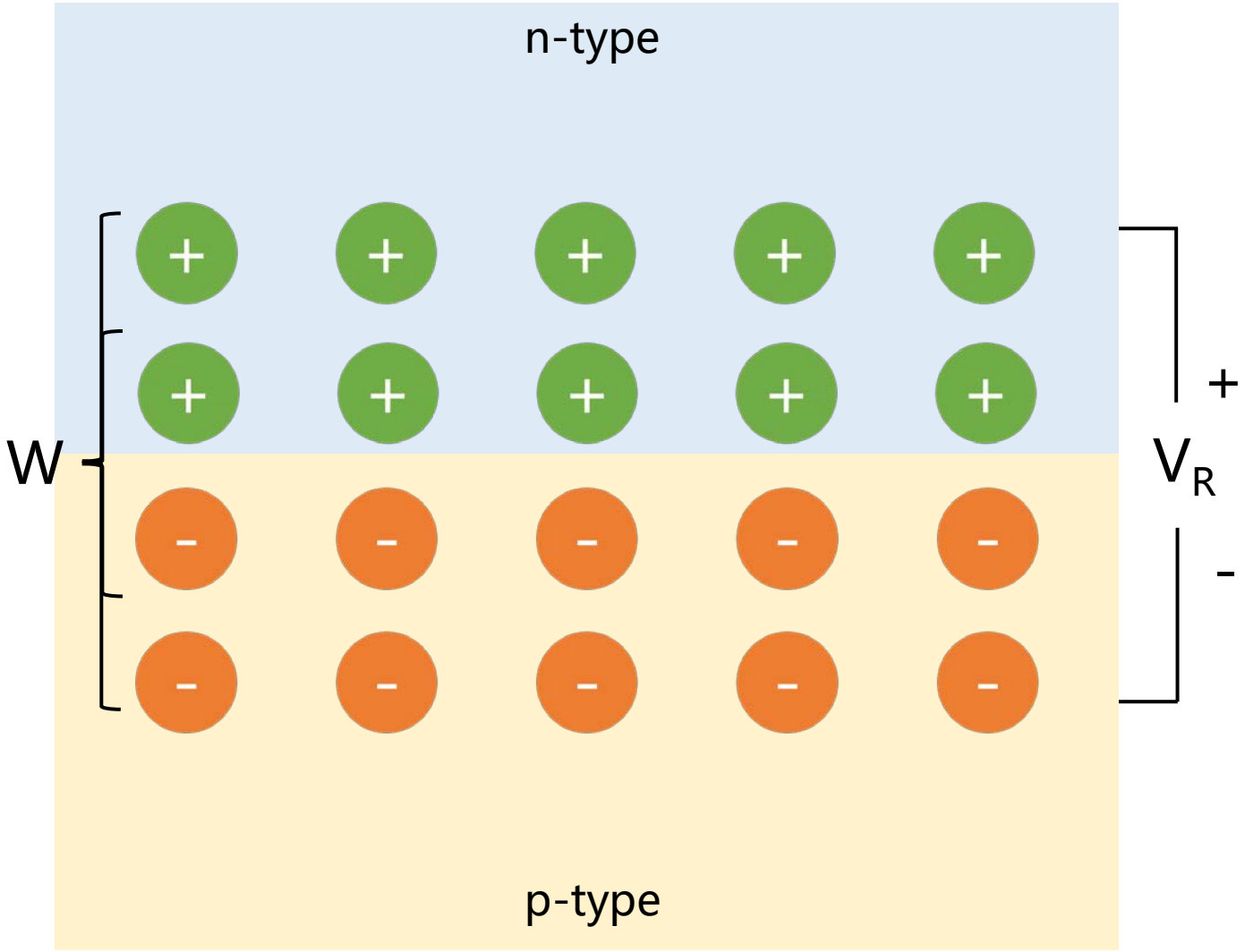
$$W = \left\{ \frac{2\epsilon_s (V_{bi} + V_R)}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{1/2}$$

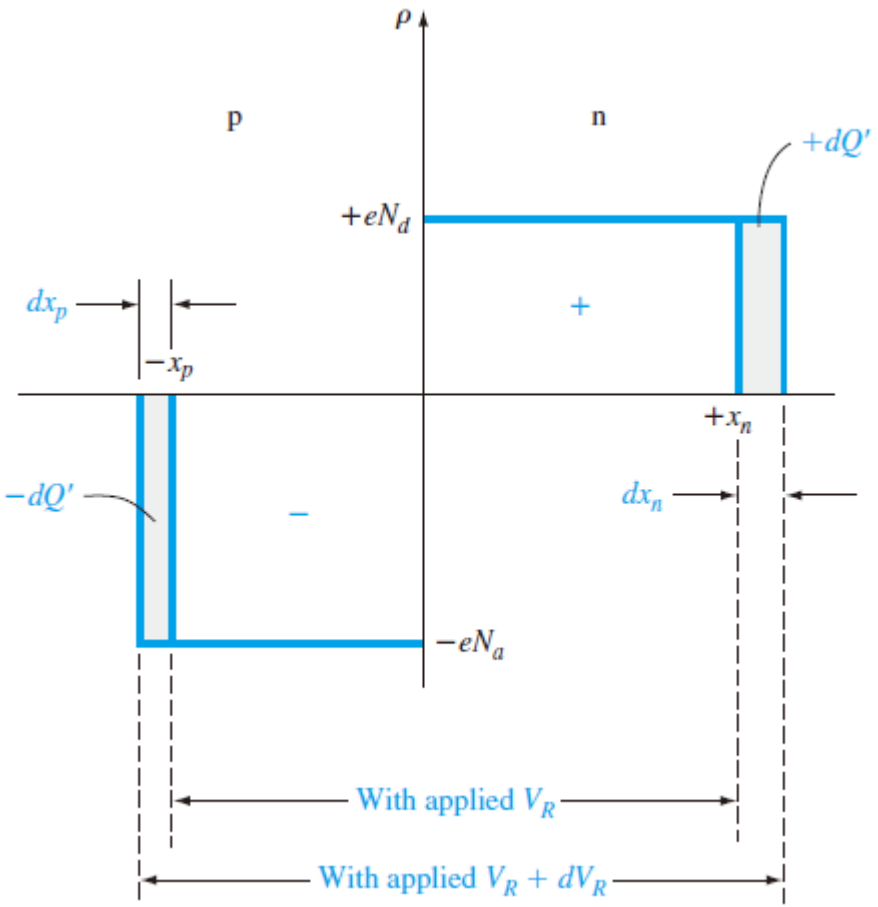
Most of the potential drop still falls in the junction region:

$$E_{max} = -\left\{ \frac{2e(V_{bi} + V_R)}{\epsilon_s} \left(\frac{N_a N_d}{N_a + N_d} \right) \right\}^{1/2}$$

$$E_{max} = \frac{-2(V_{bi} + V_R)}{W}$$

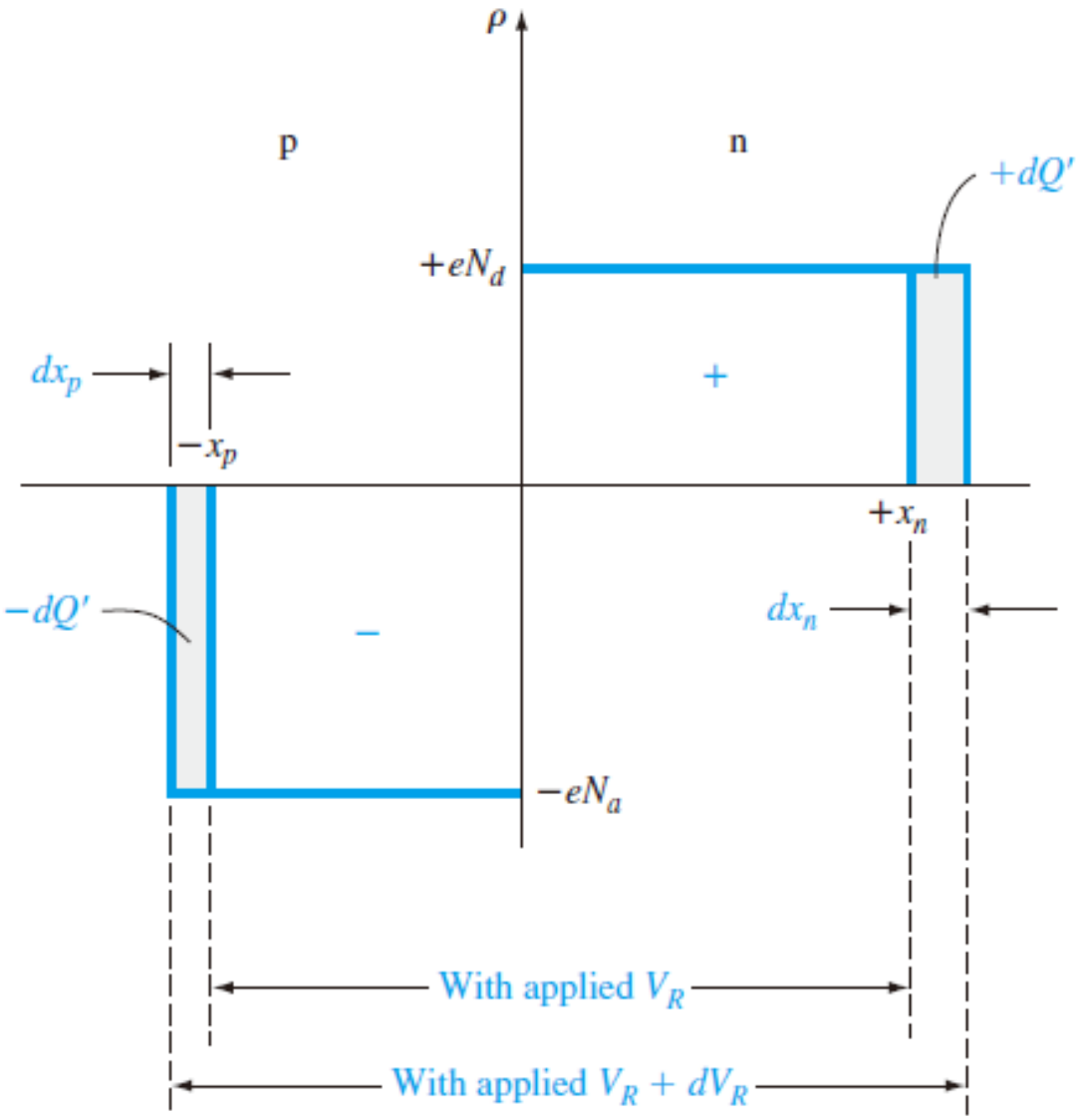
The physics of pn junction: junction capacitance





pn junction has capacitance

The physics of pn junction: junction capacitance



$$C' = \frac{dQ'}{dV_R} \qquad dQ' = eN_d\,dx_n = eN_a\,dx_p$$

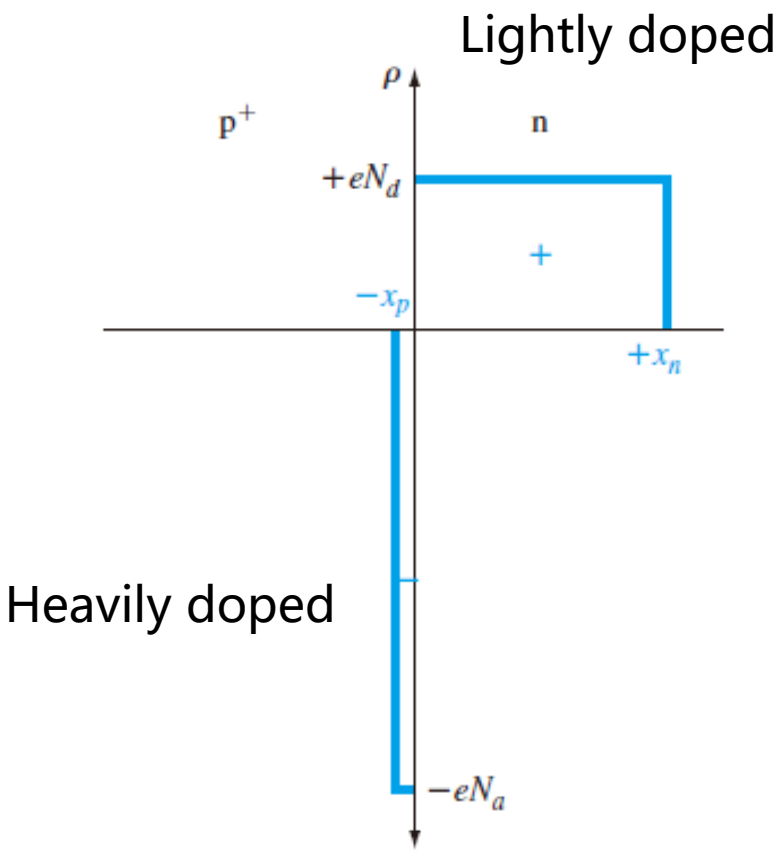
$$x_n = \left\{\frac{2\epsilon_s(V_{bi} + V_R)}{e}\left[\frac{N_a}{N_d}\right]\left[\frac{1}{N_a + N_d}\right]\right\}^{1/2}$$

$$C' = \frac{dQ'}{dV_R} = eN_d\frac{dx_n}{dV_R}$$

$$\boxed{C' = \left\{\frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right\}^{1/2}} = \frac{\epsilon_s}{W}$$

The capacitance of pn junction is the same as a parallel plate capacitor, the distance between the two plates is the depletion region width W.

The physics of pn junction: one-sided junction



In this p^+n case, the depletion region width is mainly determined by the doping concentration of the lightly doped semiconductor:

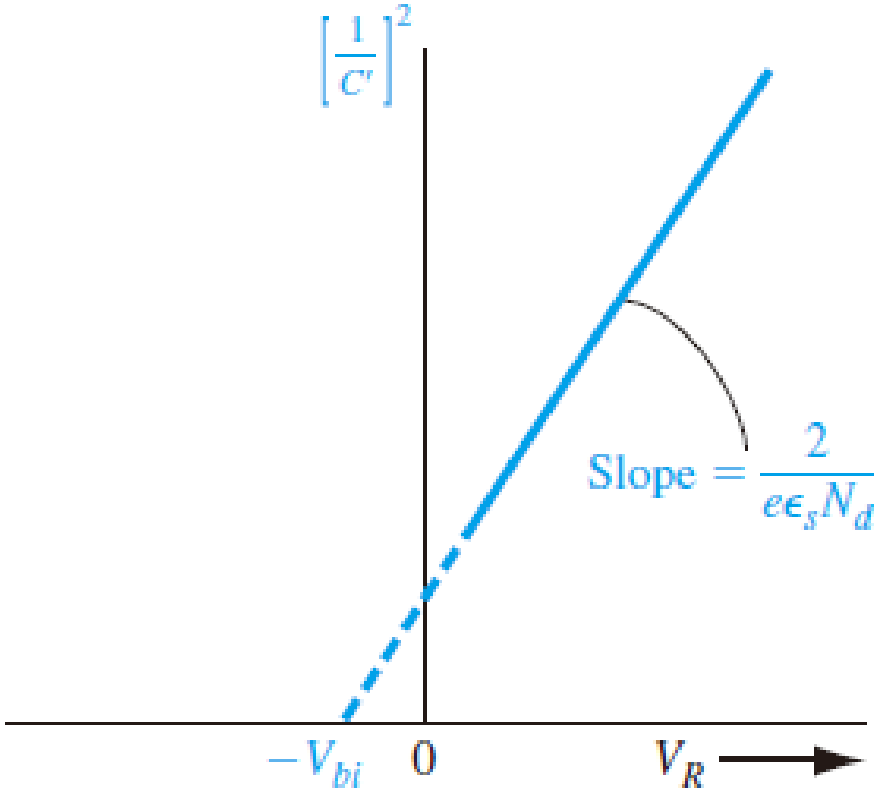
$$W \approx \left\{ \frac{2\epsilon_s (V_{bi} + V_R)}{eN_d} \right\}^{1/2}$$

$$C' \approx \left\{ \frac{e\epsilon_s N_d}{2(V_{bi} + V_R)} \right\}^{1/2}$$

The physics of pn junction: one-sided junction

A useful method to find out the doping concentration:

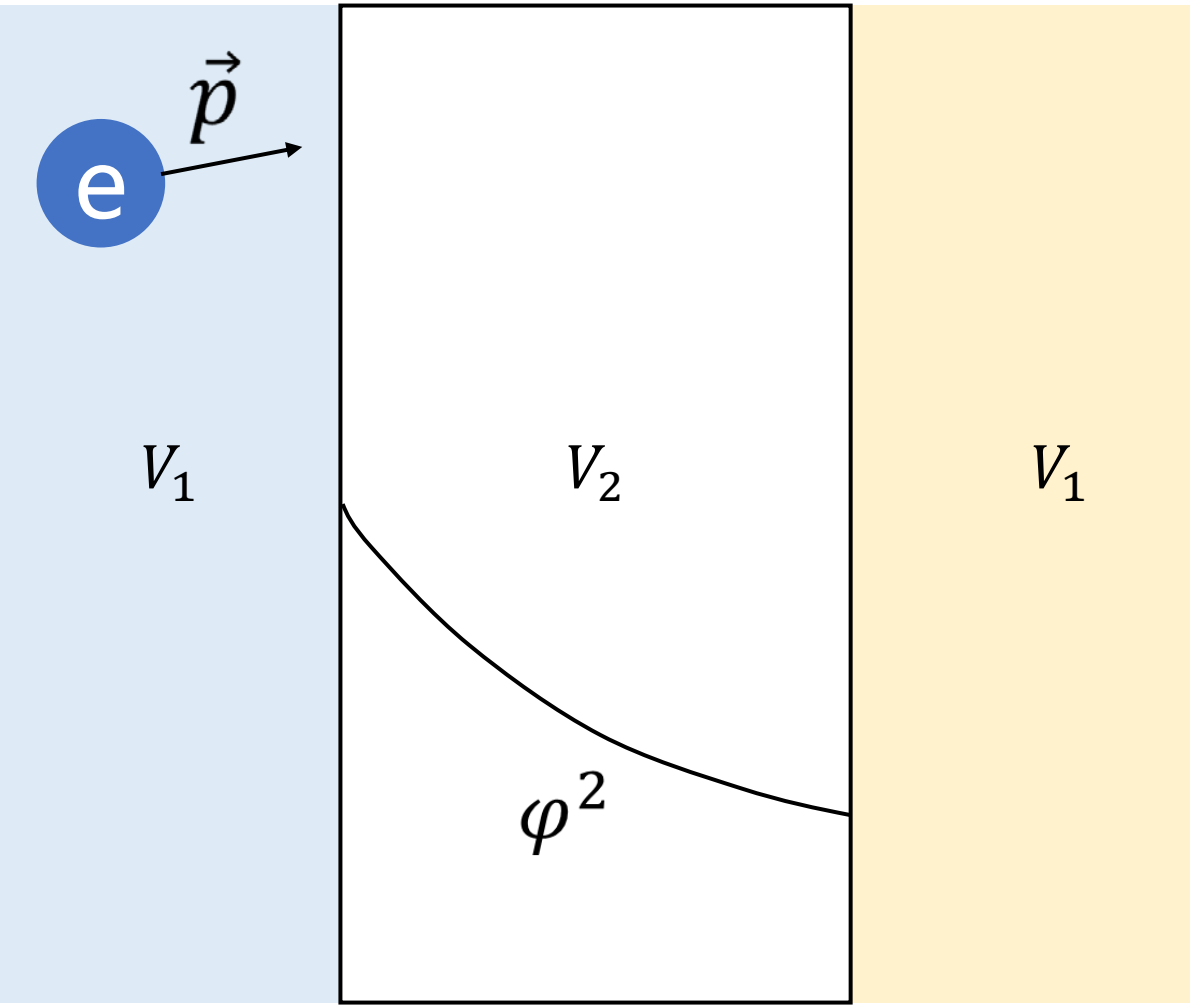
$$C' \approx \left\{\frac{e\epsilon_s N_d}{2(V_{bi}+V_R)}\right\}^{1/2} \qquad \left(\frac{1}{C'}\right)^2 = \frac{2(V_{bi}+V_R)}{e\epsilon_s N_d}$$



1. Measure capacitance-voltage curve with a suitable frequency;
2. Calculate the differentiate capacitance C';
3. Draw $(\frac{1}{C'})^2 - V$;
4. Calculate N_d from its slope.

The physics of pn junction: junction break down

Mechanism 1: Zener effect (Tunneling)



$$eV_1 + \frac{{p_1}^2}{2m} = eV_2 + \frac{{p_2}^2}{2m}$$

What if $eV_2 \gg eV_1 + \frac{{p_1}^2}{2m}$?

Then $\frac{{p_2}^2}{2m}$ must be less than zero

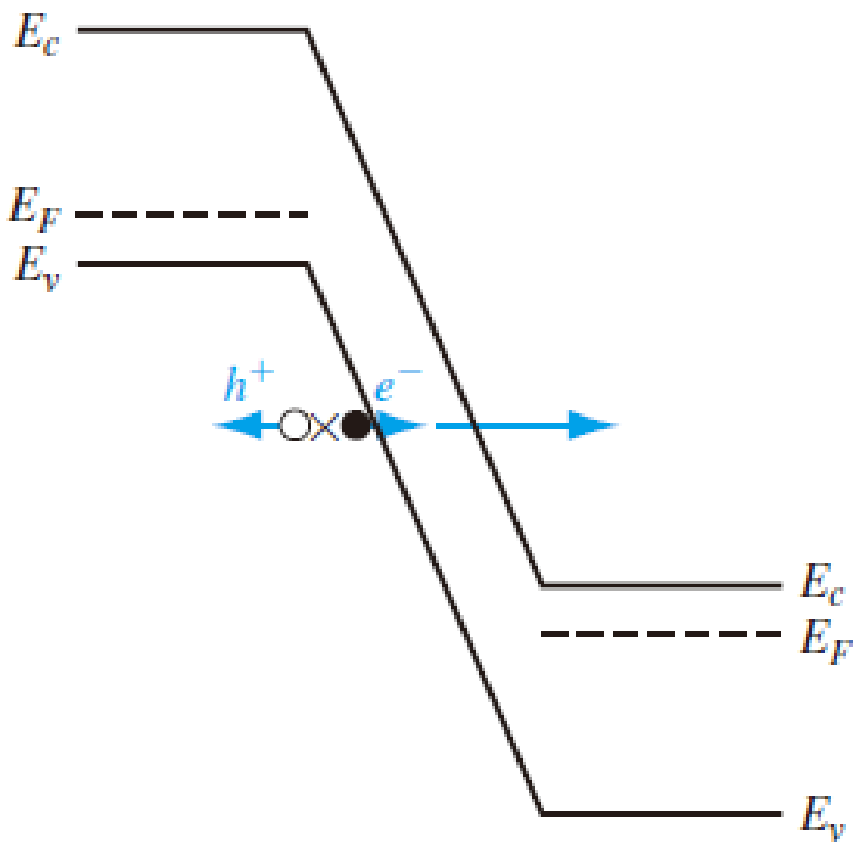
$$p_2 = ip$$

The wave function of electron in region V_2 is:

$$\varphi = e^{-\frac{px}{\hbar}}$$

The physics of pn junction: junction break down

Mechanism 1: Zener effect (Tunneling)

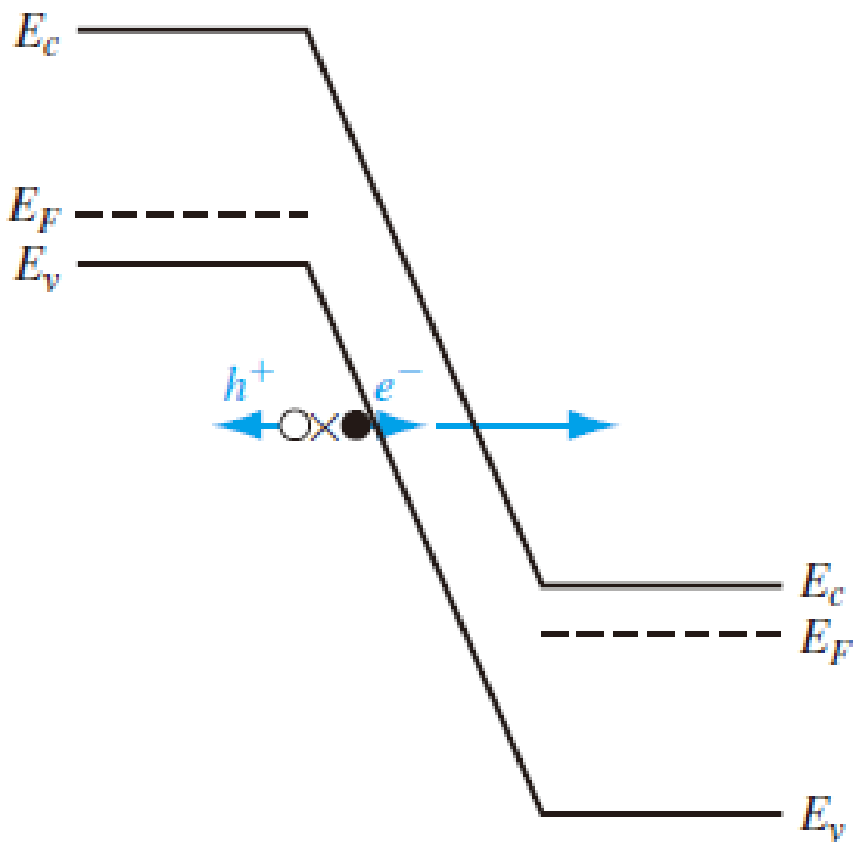


If the junction is narrow, electrons can tunnel through the depletion region of the pn junction.

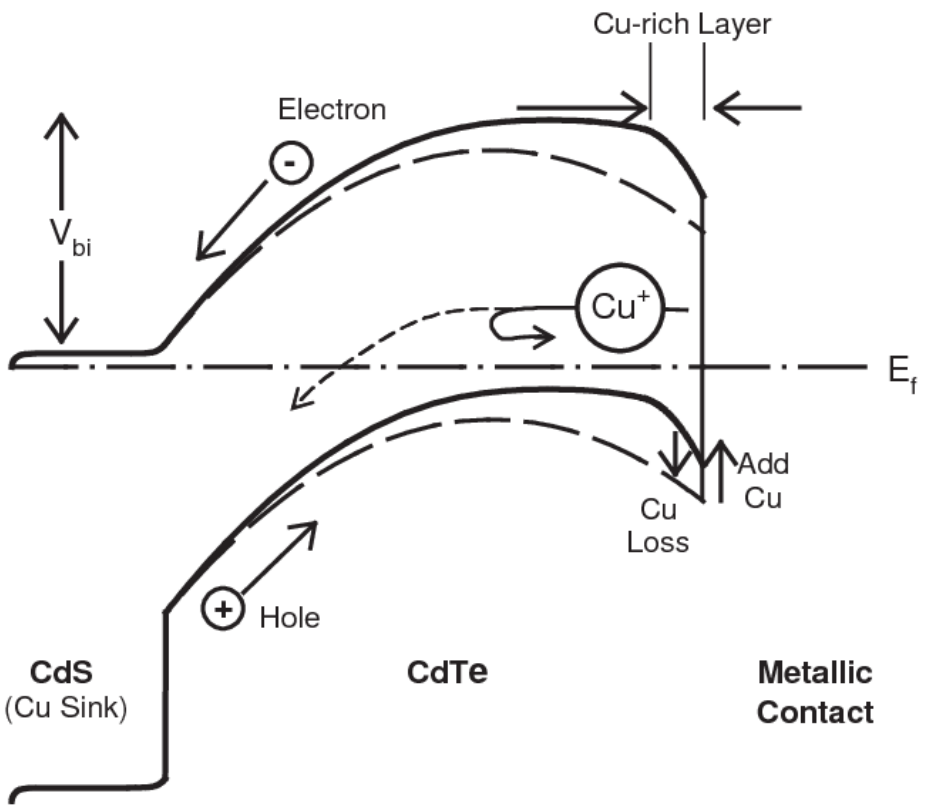
Narrow junction means the junction must be heavily doped.

The physics of pn junction: junction break down

Mechanism 1: Zener effect (Tunneling)

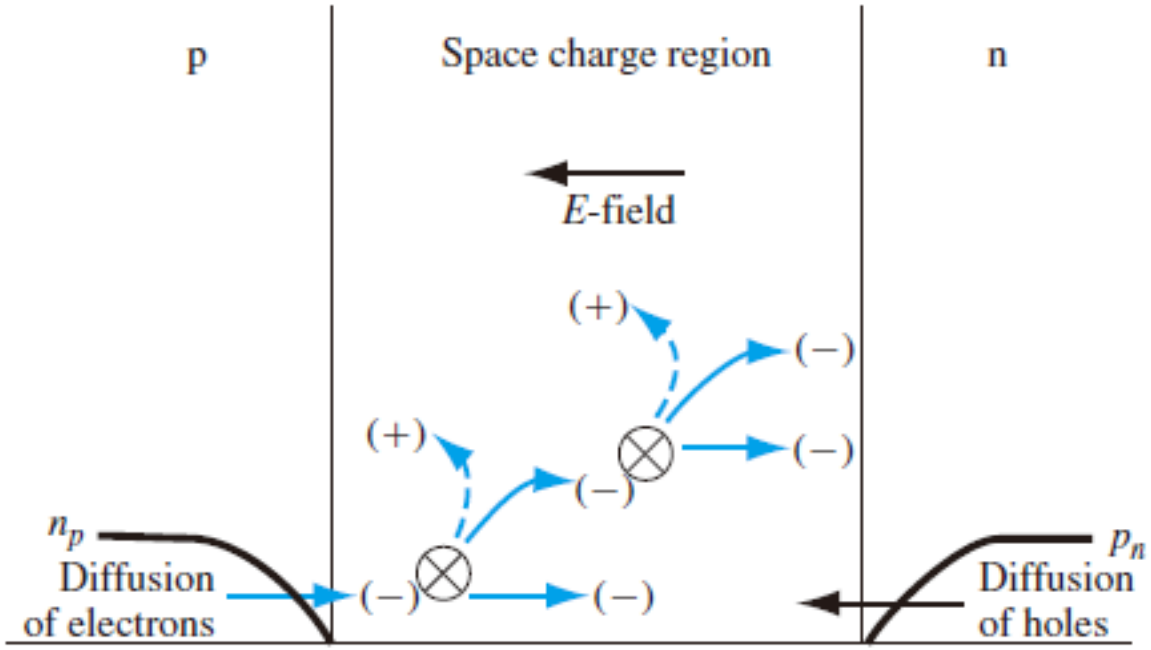


This is how you deal with carrier collection of real semiconductors



The physics of pn junction: junction break down

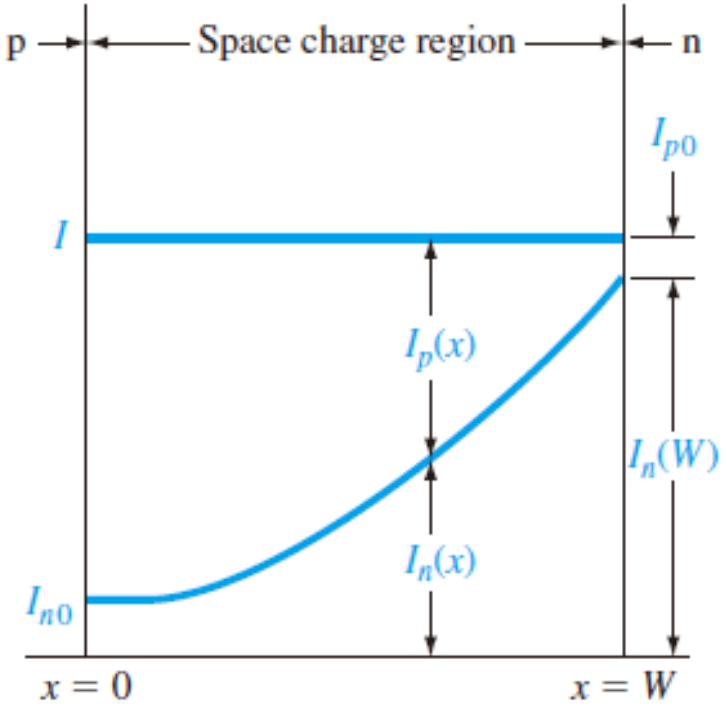
Mechanism 2: avalanche breakdown



1. The electric field is too strong so that one electron gets enough energy to break the chemical bonds of the semiconductor ;
2. One broken bond releases two electrons, thus creates a chain reaction in the depletion region.

The physics of pn junction: junction break down

Mechanism 2: avalanche breakdown



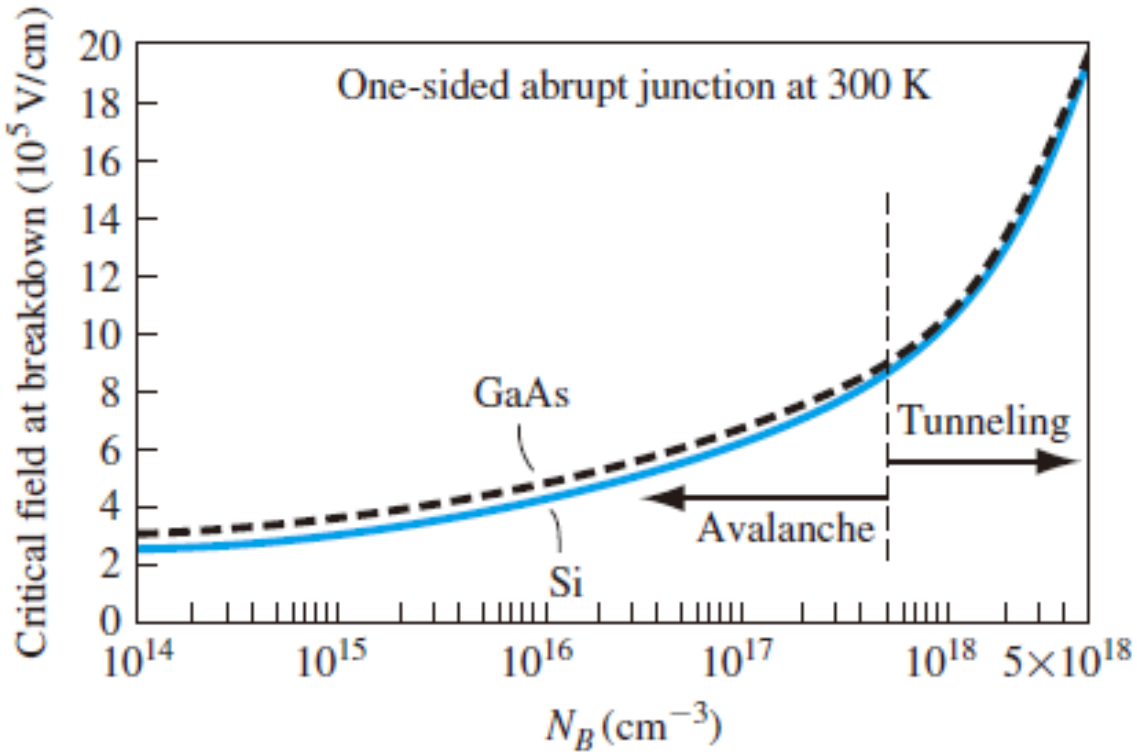
Consider a p^+n junction, the critical electric field before avalanche break down is:

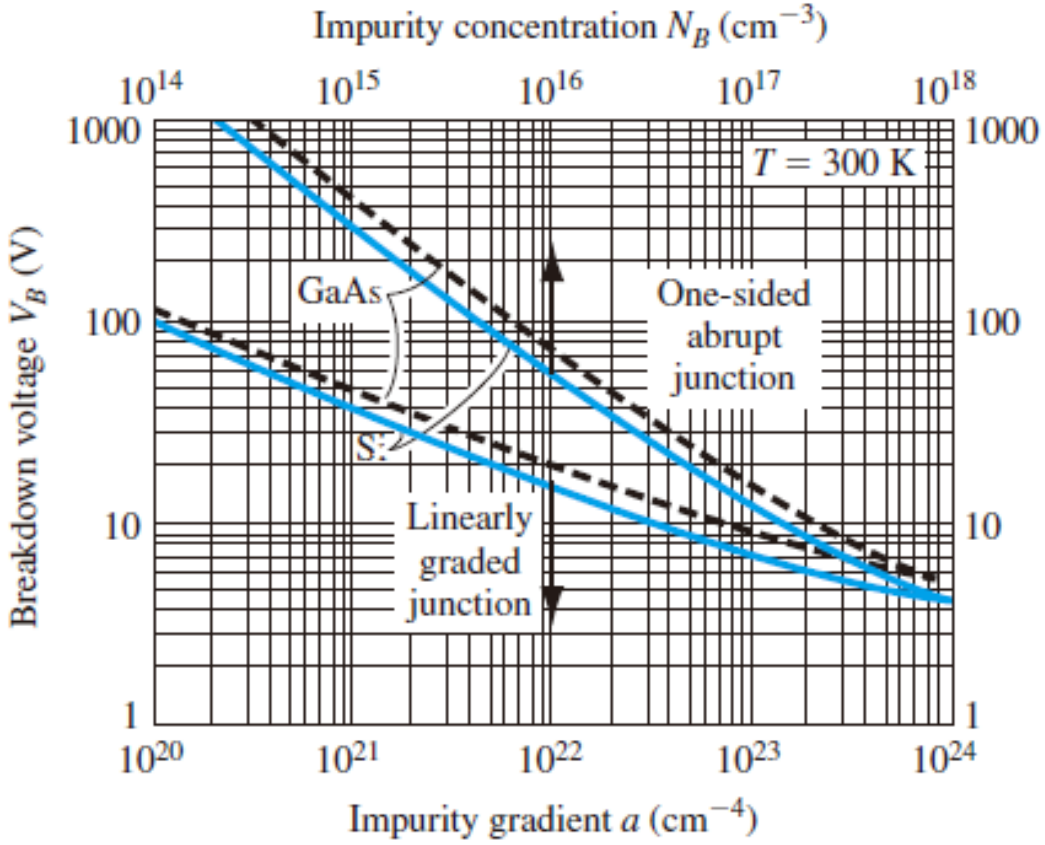
$$E_{max} = \frac{eN_d x_n}{\epsilon_s}$$

$$x_n \approx \left\{ \frac{2\epsilon_s V_R}{e} \cdot \frac{1}{N_d} \right\}^{1/2}$$

$$\boxed{V_B = \frac{\epsilon_s E_{crit}^2}{2eN_B}}$$

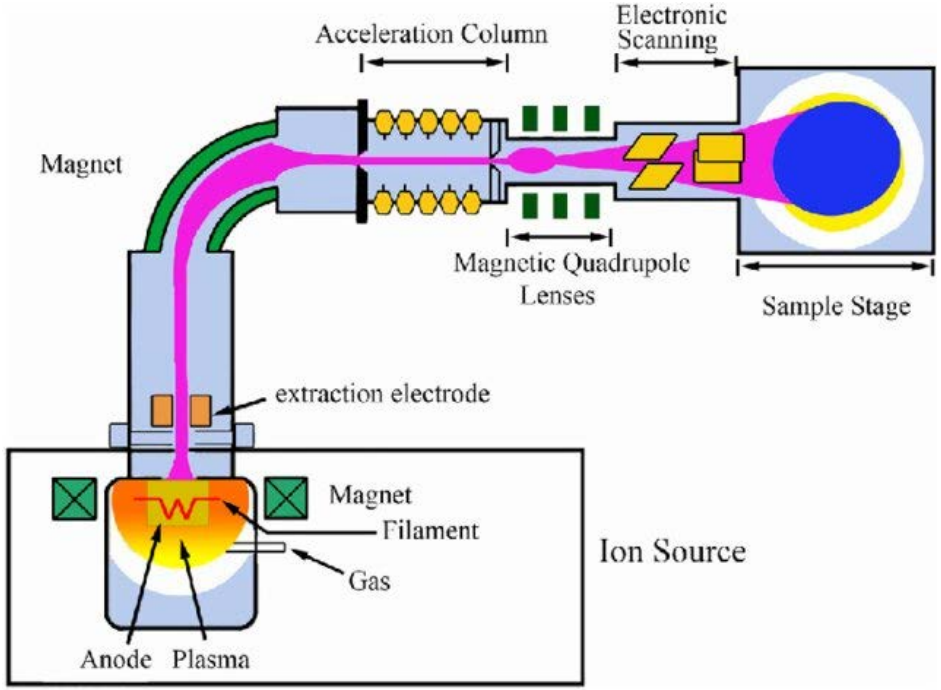
The physics of pn junction: junction break down

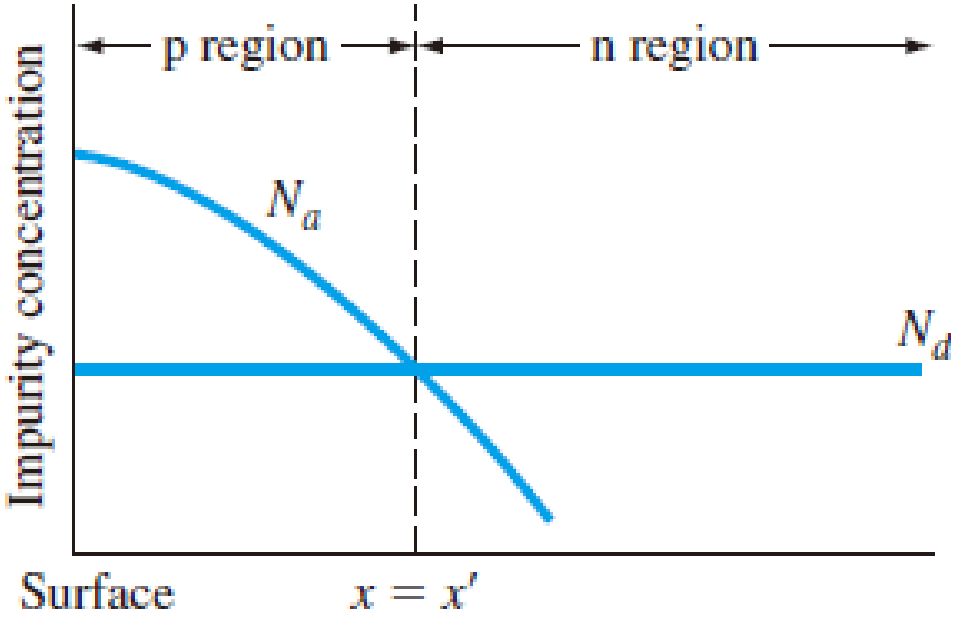




Nonuniformly doped junctions

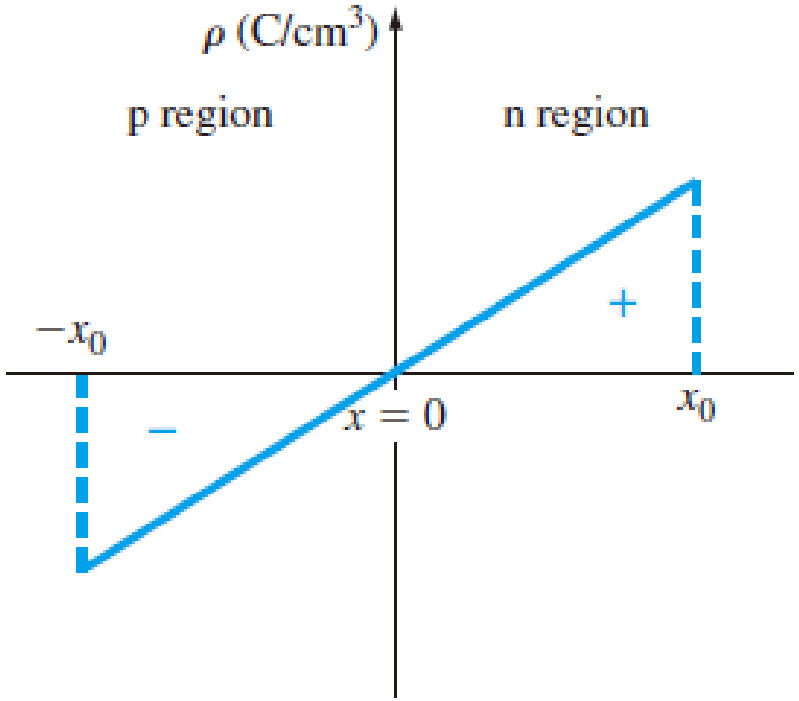
Linearly Graded Junctions





Nonuniformly doped junctions

Linearly Graded Junctions



Charge density

$$\rho(x) = eax$$

$$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_s} = \frac{eax}{\epsilon_s}$$

$$E = \int \frac{eax}{\epsilon_s}\,dx = \frac{ea}{2\epsilon_s}\left(x^2 - x_0^2\right)$$

$$\phi(x) = \frac{-ea}{2\epsilon_s}\left(\frac{x^3}{3} - x_0^2 x\right) + \frac{ea}{3\epsilon_s}x_0^3$$

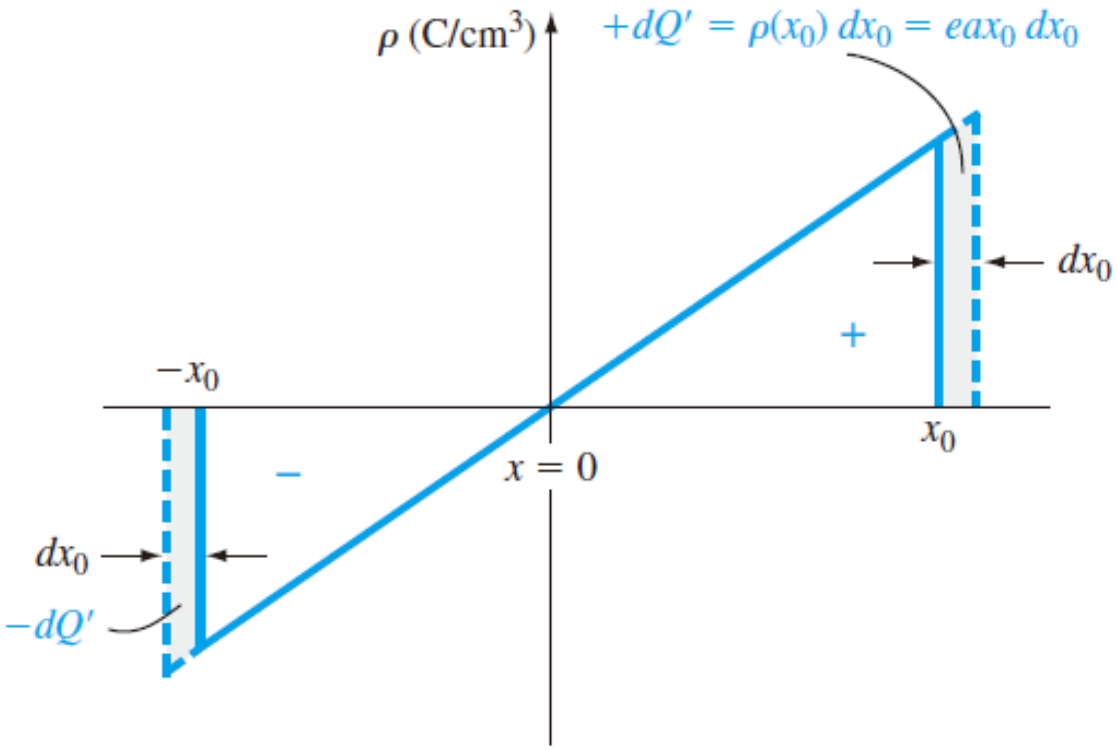
$$V_{bi} = \phi(x_0) = V_t \ln\left[\frac{N_d(x_0)N_a(-x_0)}{n_i^2}\right]$$

$$= V_t \ln\left(\frac{ax_0}{n_i}\right)^2$$

$$x_0 = \left\{\frac{3}{2}\cdot\frac{\epsilon_s}{ea}(V_{bi} + V_R)\right\}^{1/3}$$

Nonuniformly doped junctions

Linearly Graded Junctions



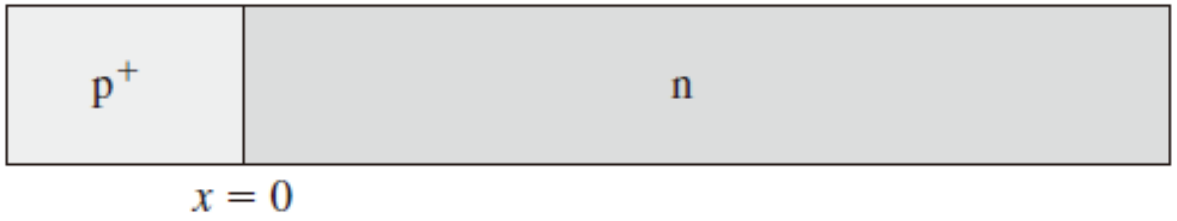
$$C' = \frac{dQ'}{dV_R} = (eax_0)\frac{dx_0}{dV_R}$$

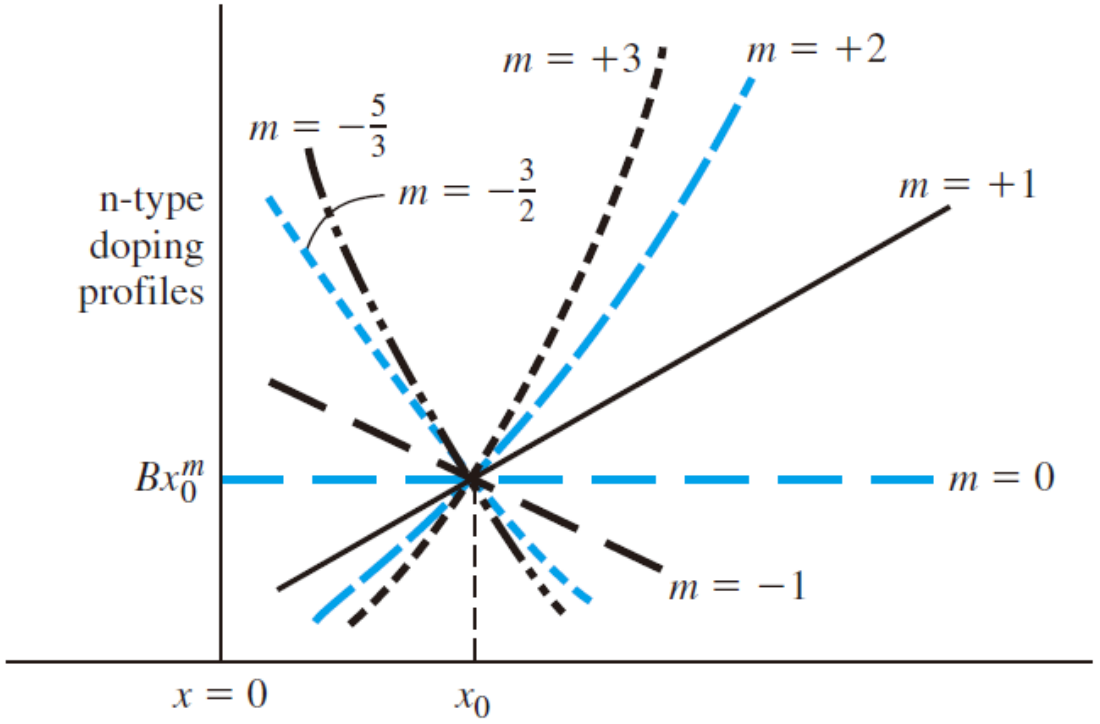
$$C' = \left\{\frac{ea\epsilon_s^2}{12(V_{bi} + V_R)}\right\}^{1/3}$$

Nonuniformly doped junctions

Hyperabrupt Junction

$$N = Bx^m$$





Hyperabrupt Junction refers to junctions with negative exponential power m

Nonuniformly doped junctions

Hyperabrupt junction

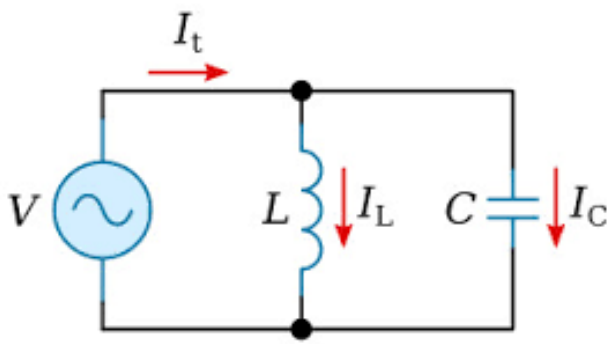
m can be determined through differentiate capacitance:

$$C' = \left\{ \frac{eB\epsilon_s^{(m+1)}}{(m+2)(V_{bi}+V_R)} \right\}^{1/(m+2)}$$

When m is negative, the capacitance becomes a very strong function of reverse-biased voltage, a desired characteristic in varactor diodes. The term varactor comes from the words variable reactor and means a device whose reactance can be varied in a controlled manner with bias voltage.

Nonuniformly doped junctions

Hyperabrupt junction



$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$C = C_0(V_{bi} + V_R)^{-1/(m+2)}$$

We would like to have: $C \alpha V^{-2}$

$$m = -\frac{3}{2}$$

Exercises

7.1，7.4，7.8，7.11，7.16，7.22